

# ELEC2134 : Circuits and Signals

## PART A ( Week 1 to Week 5)

### **Transform Methods**

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

### **System Analysis**

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- Transient analysis of 1<sup>st</sup> and 2<sup>nd</sup> systems
- Circuit analysis using transform methods

## PART B (Week 7 to Week 10)

### **AC Circuit Analysis**

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- Series and parallel resonance
- Transformers and coupled coils
- Application of network theorems: Two port networks

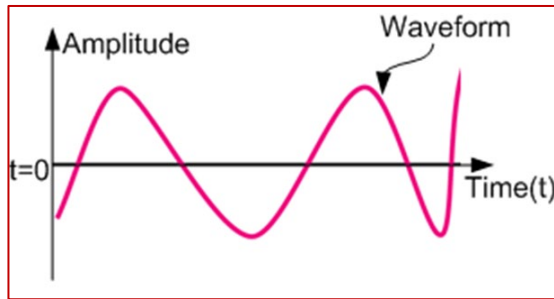
# ELEC2134 : Circuits and Signals (PART A)

## Topic 1: Transform Methods - Periodic signals

# Waveforms

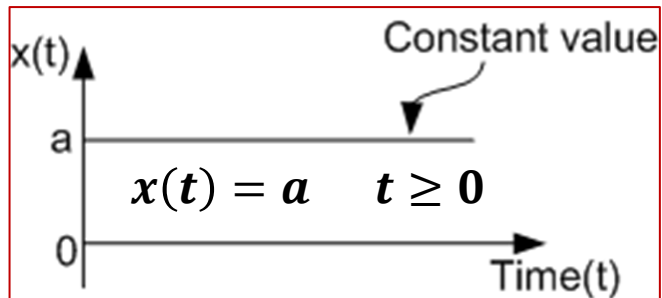
## What is a Waveform?

- A **waveform** describes the shape of a signal as it changes over time, meaning its amplitude varies with time.



## What is a Signal?

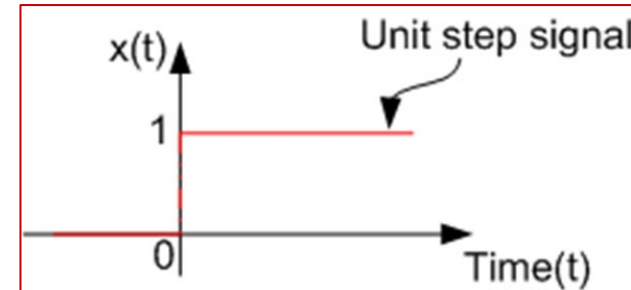
- A **signal** is a function that varies with time. When plotted against time, it forms a waveform.



## What is a Unit Step Signal?

A unit step signal is the derivative of a ramp signal

- A unit step signal stays at 0 before a certain time and suddenly becomes 1 after that time.



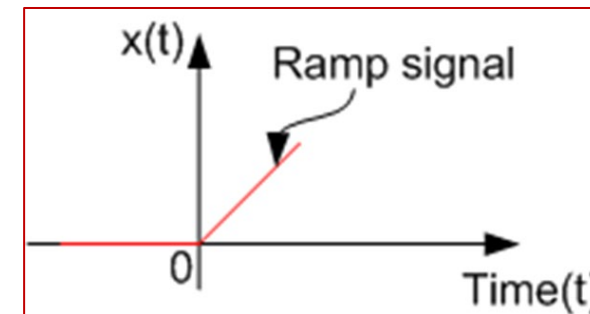
$$x(t) = 1 \quad t \geq 0$$

$$x(t) = 0 \quad t < 0$$

## What is a Ramp Signal?

A ramp signal is the integral of a unit step signal,

- A ramp signal increases linearly with time starting from zero. Its value grows like a straight line.



$$x(t) = t \quad t \geq 0$$

$$x(t) = 0 \quad t < 0$$



# Periodic Waveforms

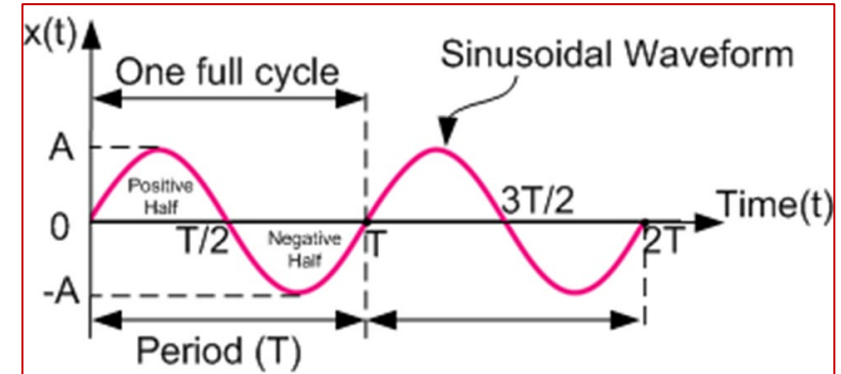
## What is a Sine Waveform?

- A sine wave (or sinusoid) is a **periodic** waveform with a smooth, continuous rise and fall, similar to the motion of a pendulum.
- The Period (T) is the time taken for the waveform to complete one full cycle and repeat itself. We define

$$\text{frequency}(f) = \frac{1}{T} \text{ Hertz (Hz)}$$

[Hertz: cycles per second, 1 Hz = 1 cycle per second]

- Units of period (T): seconds (s), millisecond (ms) or microsecond ( $\mu\text{s}$ ) etc.



Peak Amplitude = A

Peak-to-peak Amplitude = 2A



**Note:** AC voltage signal is a sinusoidal waveform that operates at a fundamental frequency of 50 Hz [Period (T) = 20 ms].

Peak Amplitude (A) =  $\sqrt{2} \times 230$  Volts

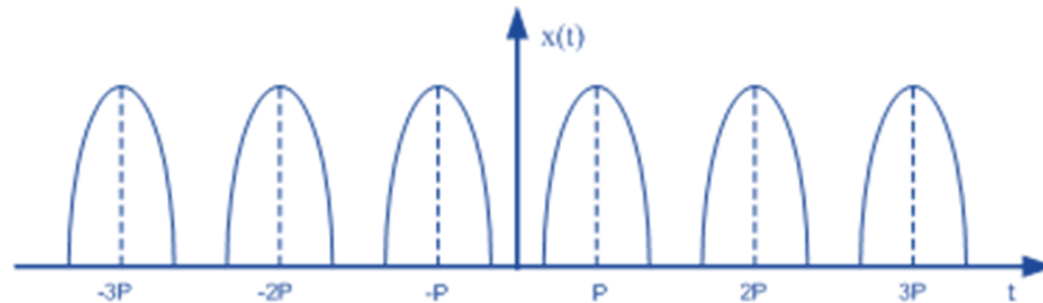
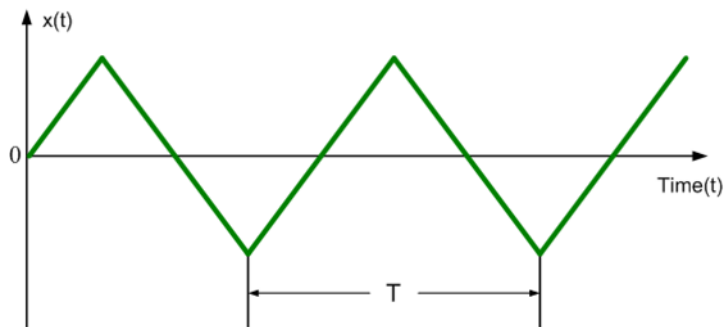
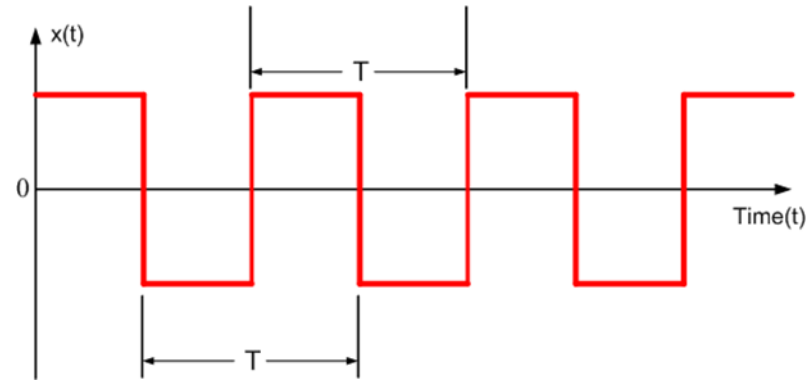
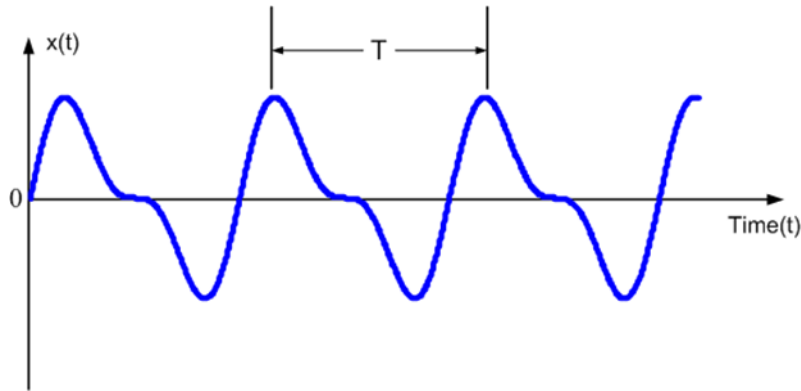
| Frequency (f)          | Period (f=1/T)                                                                | Note                         |
|------------------------|-------------------------------------------------------------------------------|------------------------------|
| f = 1 kHz (kilo Hertz) | $T = 1/f = 1/1000 = 1 \times 10^{-3} \text{ s} = 1 \text{ ms}$                | ms: milliseconds             |
| f = 1 MHz (Mega Hertz) | $T = 1/f = 1/(1 \times 10^6) = 1 \times 10^{-6} \text{ s} = 1 \mu\text{s}$    | $\mu\text{s}$ : microseconds |
| f = 1 GHz (Giga Hertz) | $T = 1/f = 1/(1 \times 10^9) = 1 \times 10^{-9} \text{ s} = 1 \text{ ns}$     | ns: nanoseconds              |
| f = 1 THz (Tera Hertz) | $T = 1/f = 1/(1 \times 10^{12}) = 1 \times 10^{-12} \text{ s} = 1 \text{ ps}$ | ps: picoseconds              |

# Periodic Waveforms (or Functions)

- Any function or waveform that satisfies

$$x(t) = x(t + T)$$

is a **periodic function** where  $T$  is the period. In general  $x(t) = x(t + nT)$ ,  $n = \pm 1, \pm 2, \pm 3, \dots$





# Sinusoidal Waveform Equation

$$x(t) = A \sin(2\pi ft) \quad \text{--- (1)}$$

Instantaneous amplitude  
(or Instantaneous value)

Peak amplitude

Frequency (Hz)

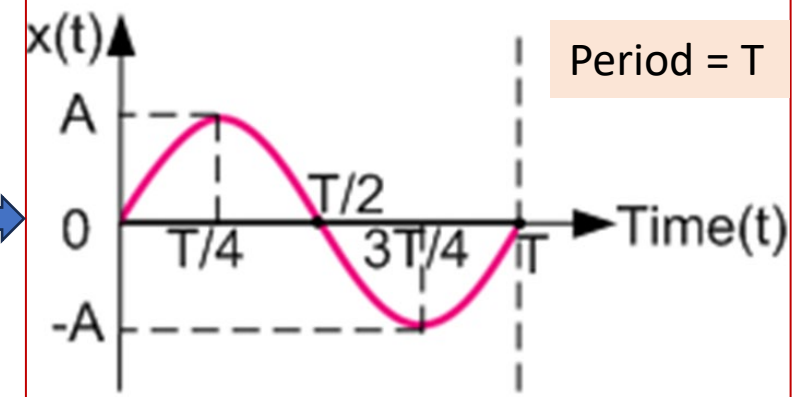
$$x(t) = A \sin\left(\frac{2\pi}{T}t\right) \quad \text{--- (2)}$$

Angular frequency ( $\omega$ ) =  $2\pi f = \frac{2\pi}{T}$  radians/sec

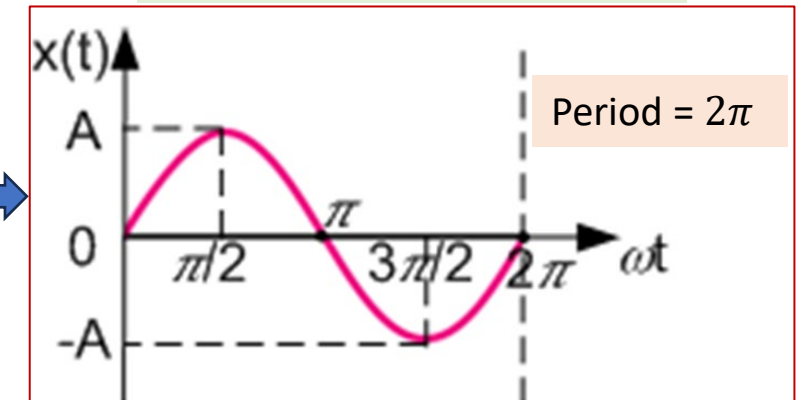
Therefore  $x(t)$  can be re-written as

$$x(t) = A \sin(\omega t) \quad \text{--- (3)}$$

$$x(t) = A \sin(2\pi ft) = A \sin\left(\frac{2\pi}{T}t\right)$$



$$x(t) = A \sin(\omega t)$$



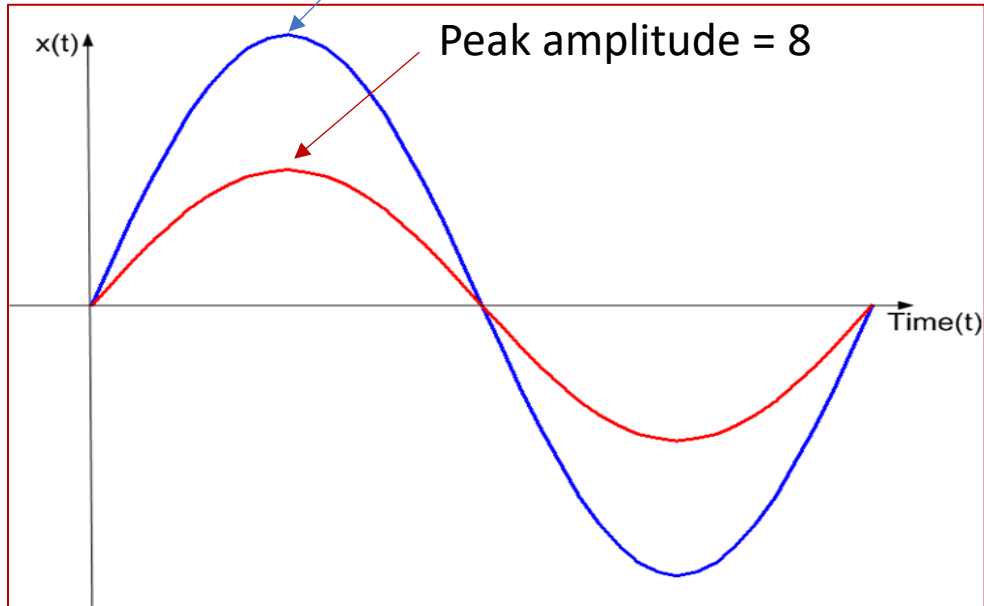
# Examples

- Sum of two sine waveforms:

$$x(t) = 10 \sin(200t) + 8 \sin(200t)$$

$$2\pi f = 200; f = \frac{200}{2\pi} \text{ Hz} = 31.83 \text{ Hz}$$

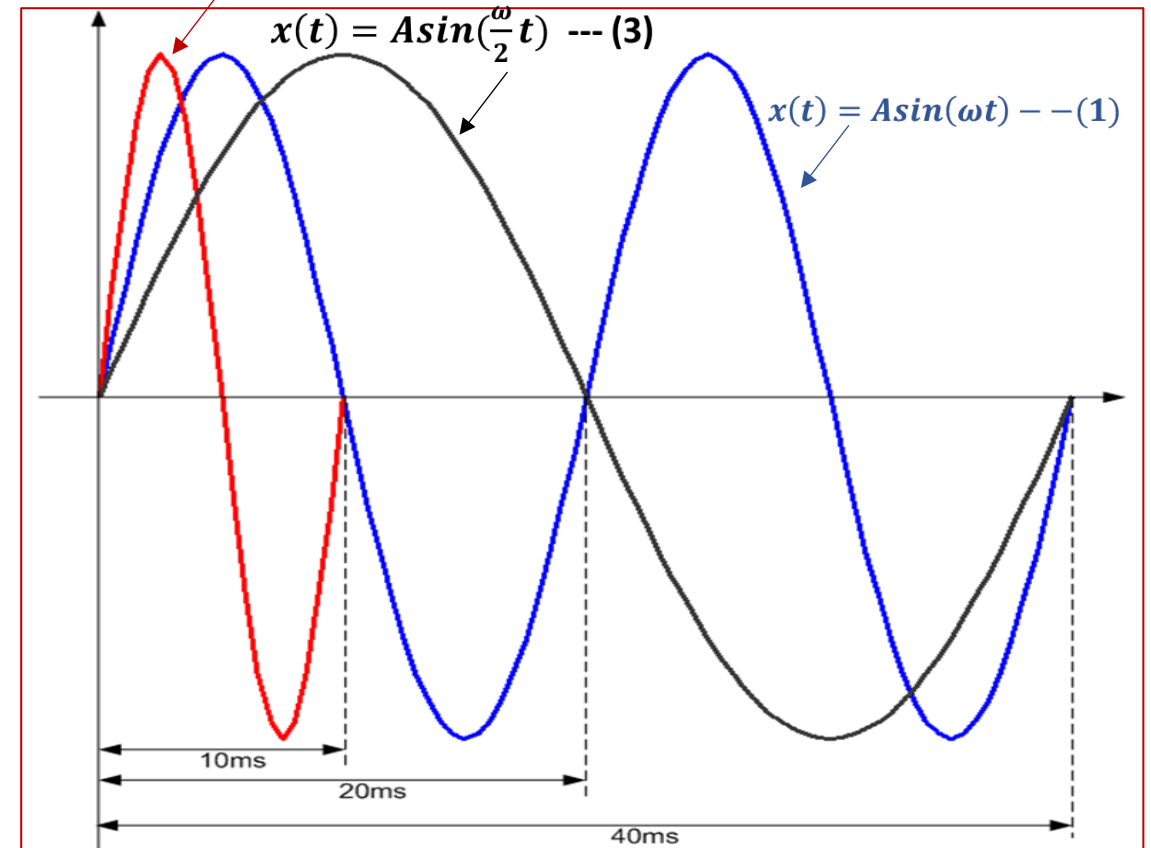
Peak amplitude = 10



Two sinusoidal signals have the same frequency but different amplitudes

- Frequency Relationships Between Signals

$$x(t) = A \sin(2\omega t) \text{ --- (2)}$$



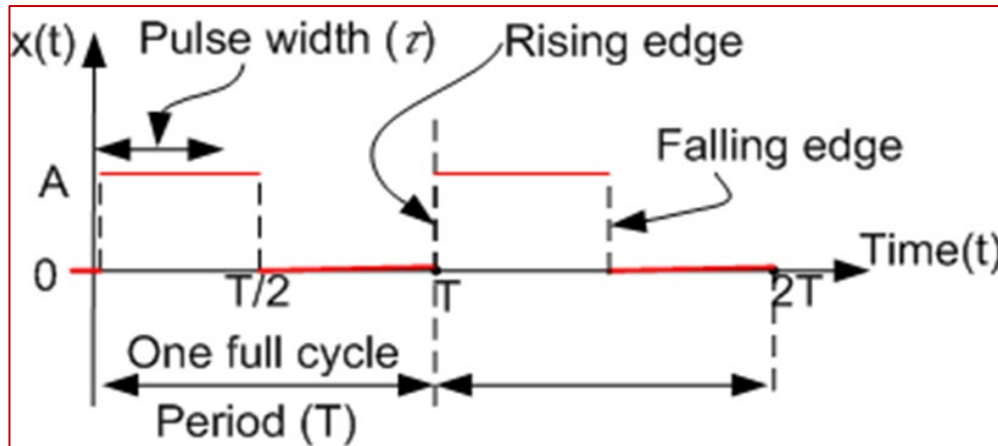
If the frequency of (1) is 50Hz, then the frequency of (2) is 100Hz and the frequency of (3) is 25 Hz



# Waveforms

## What is a Square Waveform?

- A square waveform is widely used in electronic circuits (e.g., computers) for clock and timing control signals. It has very steep vertical transitions (rising and falling edges) and a flat top and flat bottom.



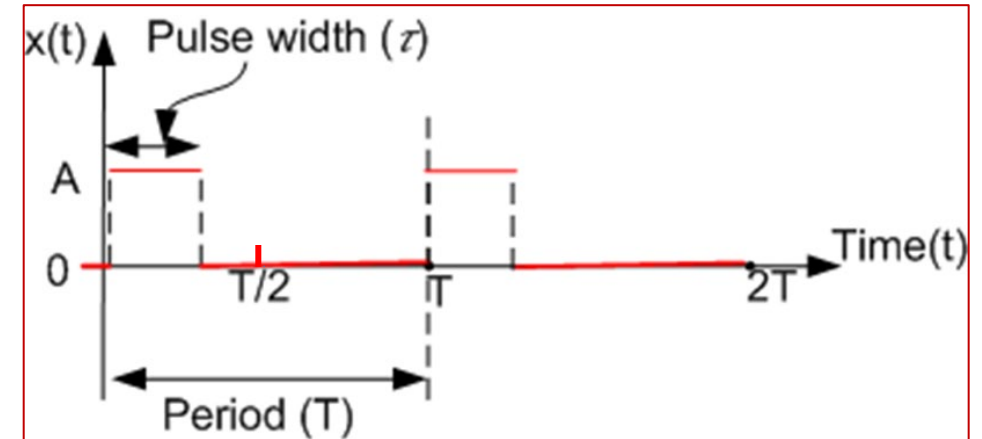
$$\text{Pulse Width } (\tau) = \frac{T}{2}; \text{ Frequency} = \frac{1}{T}$$

**Duty Cycle = 50%**

**Note:** A square wave consists of many sine waves with different frequencies and amplitudes.

## What is a Rectangular Waveform?

- A rectangular waveform resembles a square wave, but the duration of its high level, called the pulse width ( $\tau$ ), is shorter than half the period ( $T/2$ ).

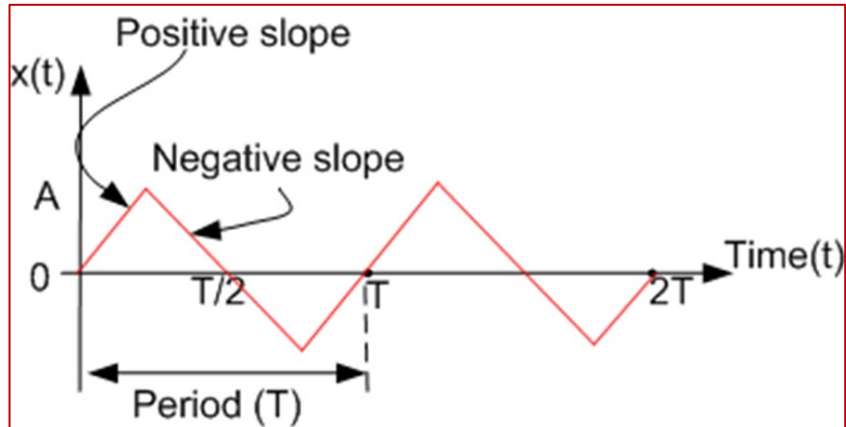


The **duty cycle** is a measure of how much time a signal is **ON** compared to the total time of one cycle.

# Waveforms

## What is a Triangular Waveform?

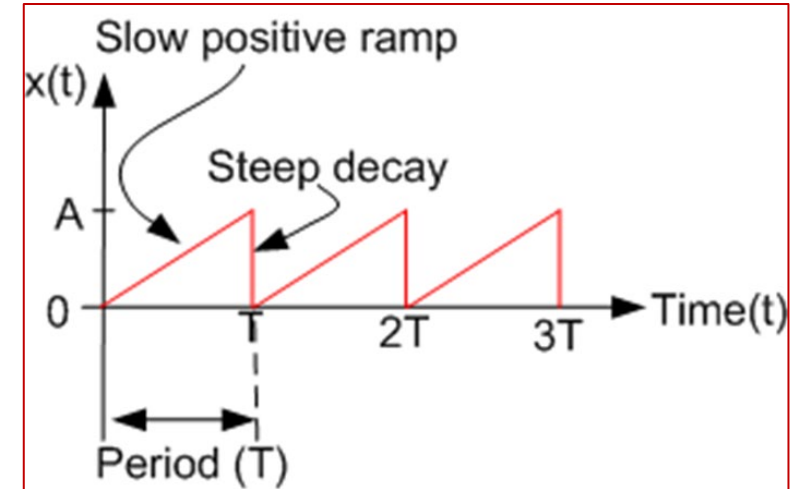
A triangular waveform is a non-sinusoidal waveform that oscillates between a positive and a negative peak value with a linear rise and fall.



**Note:** A Triangular wave consists of many sine or cosine waves with different frequencies and amplitudes.

## What is a Sawtooth Waveform?

- A sawtooth waveform is a periodic waveform whose shape resembles the teeth of a saw blade



**Note:** A Sawtooth wave consists of many sine or cosine waves with different frequencies and amplitudes.

We are not learning these waveforms for their own sake.  
We learn to recognise their structure so we can analyse more complicated signals.



# Harmonic Components

## What is a harmonic?

- If  $\omega_o = 2\pi f_o = \frac{2\pi}{T}$  is the fundamental angular frequency (rad/s) (or  $f_o$  is the frequency in Hz). Then the  $n^{th}$  harmonics is defined by
$$\omega_n = n\omega_o \quad \text{or} \quad f_n = nf_o \quad \text{where } n = 1, 2, 3, 4, \dots$$
- **Example:** If a particular AC power system has a fundamental frequency of **50Hz**, calculate the frequencies of the following harmonics:

$2^{nd}$  Harmonic =  $2 \times 50 = 100\text{Hz}$

$3^{rd}$  Harmonic =  $3 \times 50 = 150\text{Hz}$

$4^{th}$  Harmonic =  $4 \times 50 = 200\text{Hz}$

**Note:** A periodic waveform can be expressed as a sum of its harmonic components.

## What is an Average value of a Waveform?

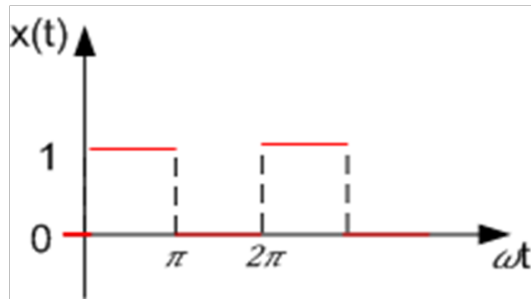
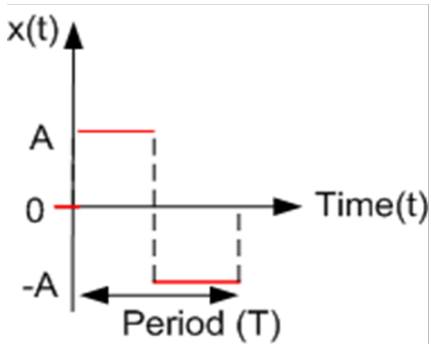
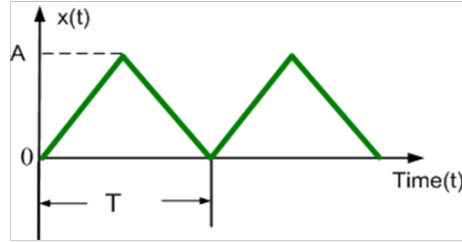
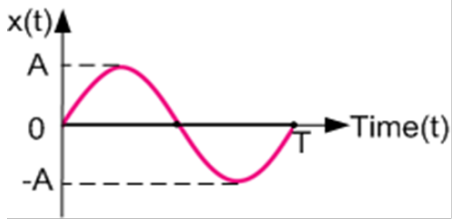
- The **average value** of a waveform is the mean (average) level of the signal over one complete cycle. It represents the overall **DC component** of the waveform.

For a periodic waveform  $x(t)$  with period  $T$ ,

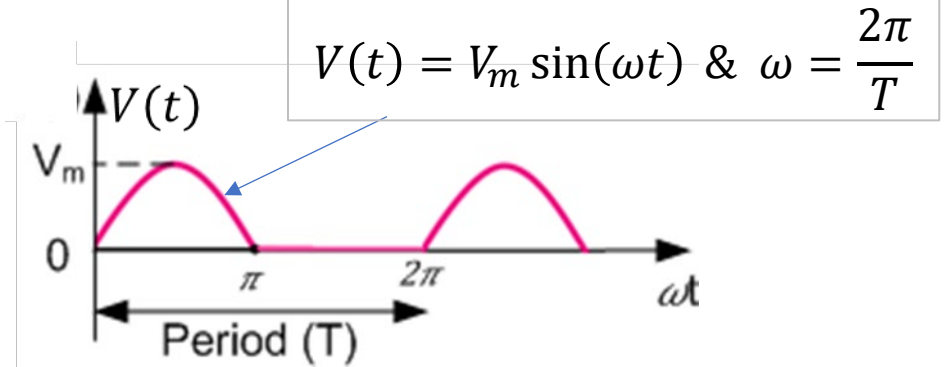
$$\text{Average Value } (V_{avg}) = \frac{1}{T} \int_0^T x(t) dt$$

# Exercises

- ✓ What is the average value for each of the following waveforms?



- ✓ Find the average value for the following waveform.

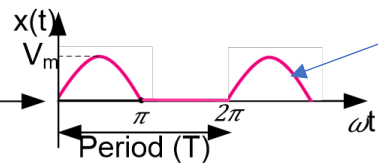


$$\begin{aligned} V_{av} &= \frac{1}{T} \int_0^T V(t) dt = \frac{1}{\frac{2\pi}{\omega}} \int_0^{\frac{2\pi}{\omega}} V_m \sin(\omega t) dt \\ &= \int_0^{\frac{\pi}{\omega}} V_m \sin(\omega t) dt + \int_{\frac{\pi}{\omega}}^{\frac{2\pi}{\omega}} 0 dt \\ &= \frac{\omega}{2\pi} \times V_m \left[ -\frac{\cos(\omega t)}{\omega} \right]_0^{\frac{\pi}{\omega}} = \frac{V_m}{\pi} \end{aligned}$$

**Note:**

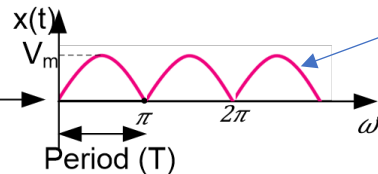
AC waveform  
(sine wave)

Half wave  
rectifier



Half-wave  
rectified  
sine wave

Full wave  
rectifier

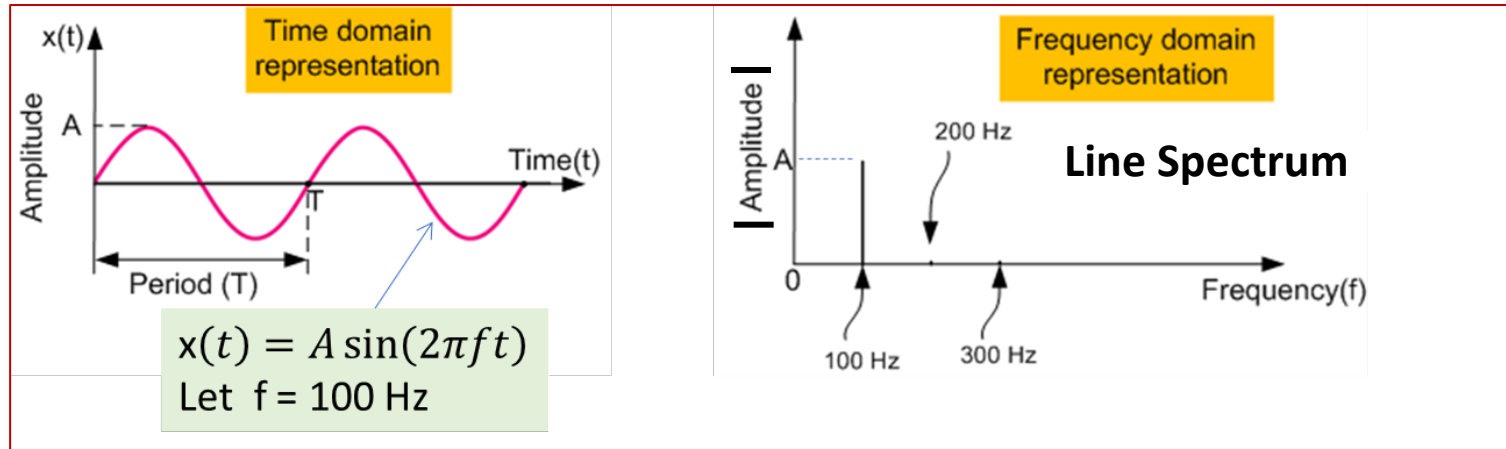


Full-wave  
rectified  
sine wave

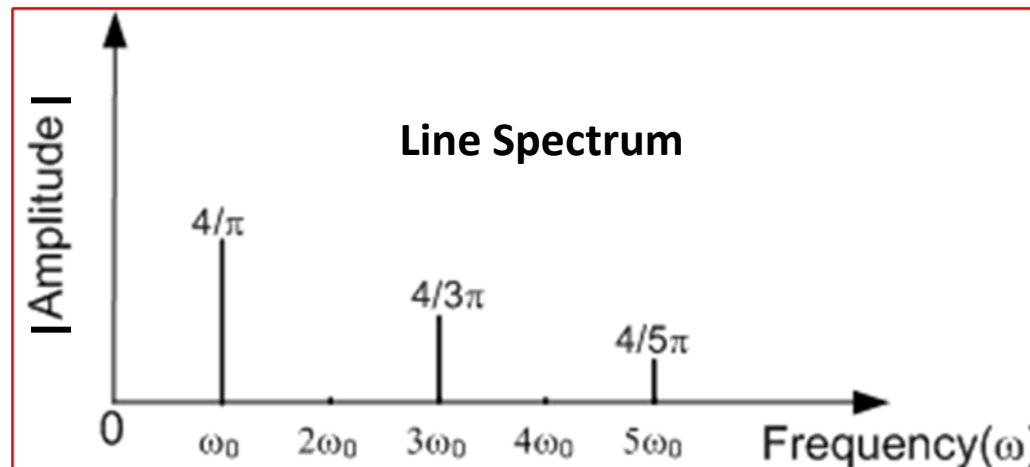


# Frequency Domain Representation of a Periodic Waveform

- The frequency domain representation shows how a time-domain signal is distributed across different frequencies



- Sketch the frequency domain representation of the time-domain signal given below:  
$$v(t) = \frac{4}{\pi} \sin(\omega_0 t) + \frac{4}{3\pi} \sin(3\omega_0 t) + \frac{4}{5\pi} \sin(5\omega_0 t) \quad [\omega_0 \text{ is the fundamental frequency}]$$



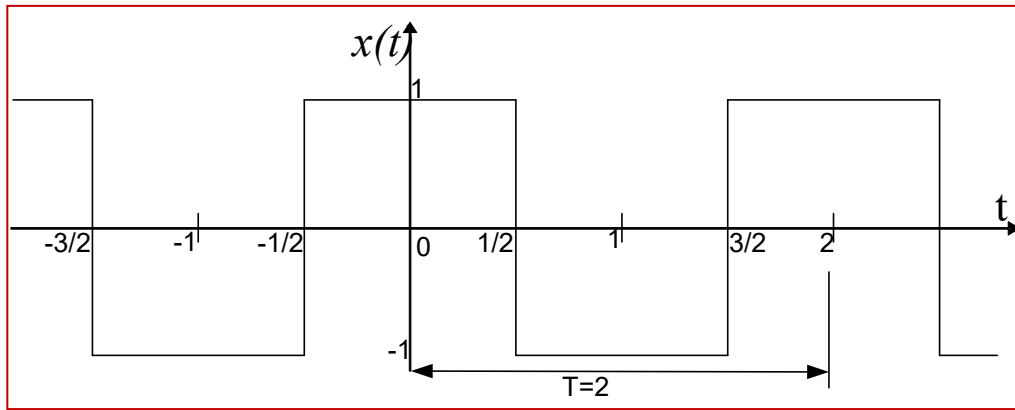
A **line spectrum** represents the magnitude of each frequency component by the length of a vertical line at the corresponding frequency.



# What is a Fourier Series?

- A Fourier series is a technique for breaking down a **periodic signal** into a sum of sinusoidal components with different frequencies and amplitudes.

Periodic Waveform



Fourier Series

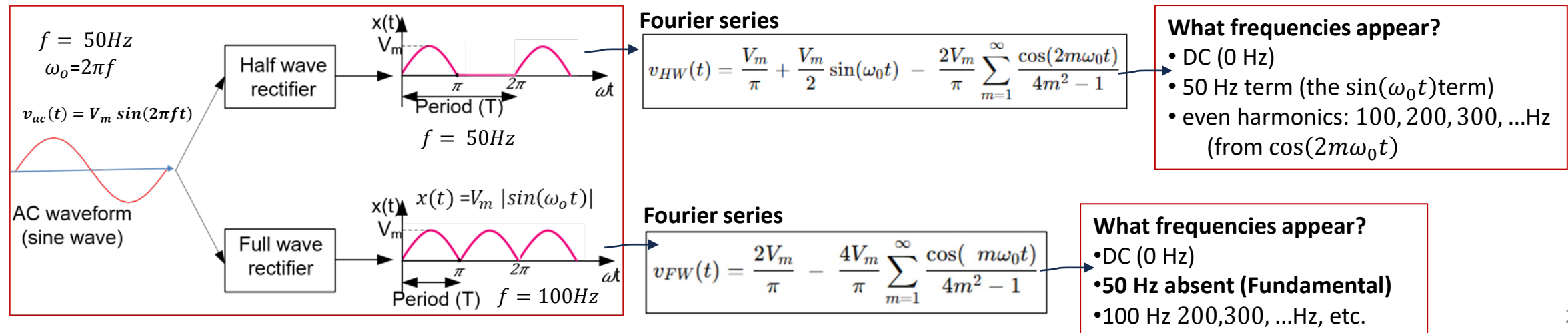
$$x(t) = \frac{4}{\pi} \cos \pi t - \frac{4}{3\pi} \cos 3\pi t + \frac{4}{5\pi} \cos 5\pi t - \frac{4}{7\pi} \cos 7\pi t + \dots$$

$$f_0 = \frac{1}{T} = \frac{1}{2} \text{ Hz}$$

The main application of the Fourier Series is to determine the harmonic components that are present in a complex periodic waveform.

# Why Do We Need to Learn Fourier Series? (Power Supply Example)

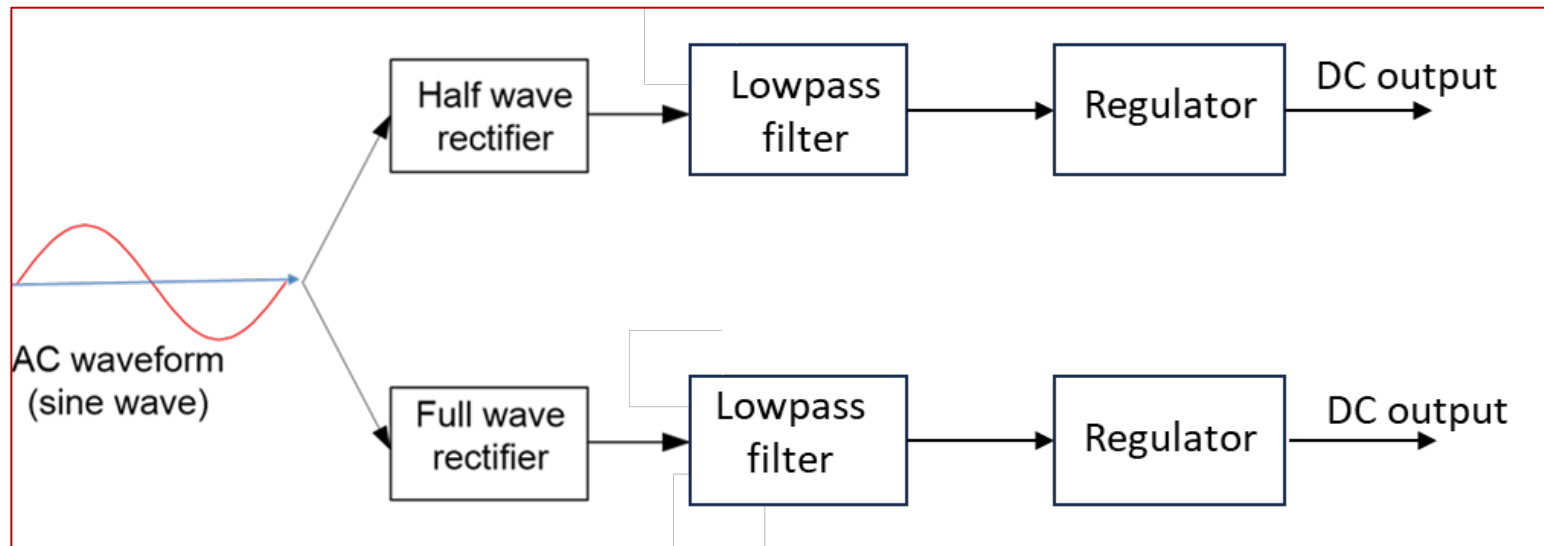
- The electrical power supplied to our homes is a 230 V sinusoidal AC voltage (frequency = 50 Hz) with an average value of zero. However, most electronic devices such as computers, mobile phones, and laboratory instruments require a DC voltage, which means a signal with a non-zero average value.
- To convert the AC voltage into DC, we use a nonlinear circuit called a **rectifier**. The rectifier changes the sinusoidal waveform so that its average value is no longer zero, thereby producing a DC component that can be used as the output of a power supply. However, the output of a rectifier is not a pure DC signal. Instead, it is a periodic but non sinusoidal waveform.
- This is where Fourier series become important. Fourier series allow us to represent any periodic waveform, including the rectifier output, as a sum of a DC component and several sinusoidal components at different frequencies. By analysing the rectifier output using Fourier series, we can determine how much DC is present and what AC frequency components remain in the signal.





# Why Do We Need to Learn Fourier Series? (Power Supply Example)

- To obtain a smoother DC voltage, the rectifier output is passed through a low pass filter, typically an RC circuit. The purpose of this filter is to pass the DC component while attenuating the unwanted sinusoidal components.
- In order to design this filter correctly, we must know which frequency components are present in the rectifier output and how strong they are. Fourier series provide exactly this information.
- Therefore, Fourier series are not just a mathematical topic. They are a practical engineering tool that helps us analyse real waveforms, design filters, and understand how power supplies and voltage regulators work in real systems.





# ELEC2134 : Circuits and Signals (PART A)

## Transform Methods:

### ▪ Fourier Series - (A) Trigonometric Fourier series

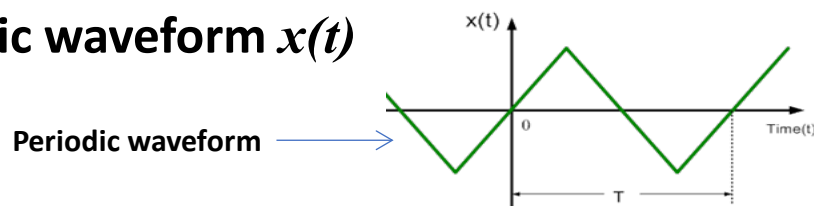
- The purpose of the trigonometric Fourier series is to represent a **periodic signal** as a sum of simple sinusoidal components (sines and cosines), each corresponding to a specific frequency

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)$$

- The coefficients  $A_n$  and  $B_n$  tell us:
  - Which frequencies are present
  - How strong each frequency component is
  - How the signal energy is distributed across frequencies

# Fourier Series : (A) Trigonometric Fourier series

- Any **periodic waveform**  $x(t)$



with a period of  $T$  seconds can be represented by an infinite series of harmonically related sinusoids. This representation is known as the trigonometric Fourier series:

$$x(t) = \underbrace{A_0}_{\text{DC}} + \underbrace{\sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)}_{\text{AC}}$$

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} B_n \sin(n\omega_0 t)$$

- The fundamental frequency of  $x(t)$  is  $\omega_0 = 2\pi f_0 = \frac{2\pi}{T}$  radians/sec. The integral multiples of  $\omega_0$  (that is  $2\omega_0, 3\omega_0, 4\omega_0$  and so on) are known as the harmonic frequencies of  $x(t)$ . Thus  $2\omega_0$  is the second harmonic,  $3\omega_0$  is the third harmonic and  $n\omega_0$  is the  $n^{\text{th}}$  harmonic of  $x(t)$ .

- The **Fourier coefficients**  $A_0, A_n$  and  $B_n$  are known as the Fourier coefficients and are calculated directly from the signal  $x(t)$  using the following equations:

$$A_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \quad \leftarrow \text{The average value of } x(t)$$

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T} \text{ radians/sec.}$$

$$A_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt$$

$$B_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt$$

$n = 1, 2, 3, \dots$

**Note** that  $x(t)$  can be rewritten as :

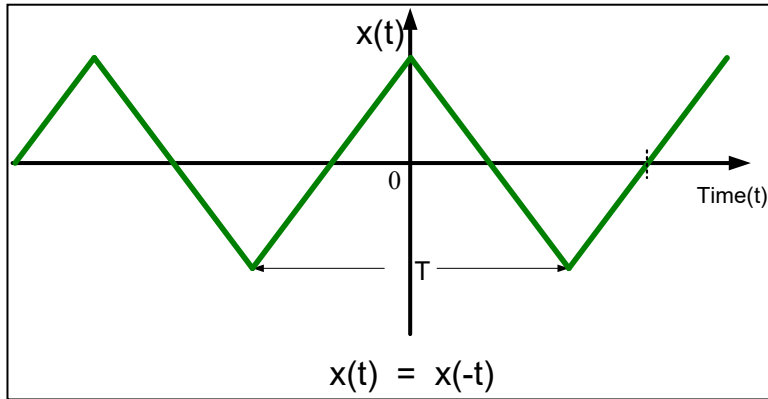
$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \cos(n\omega_0 t - 90^\circ)$$



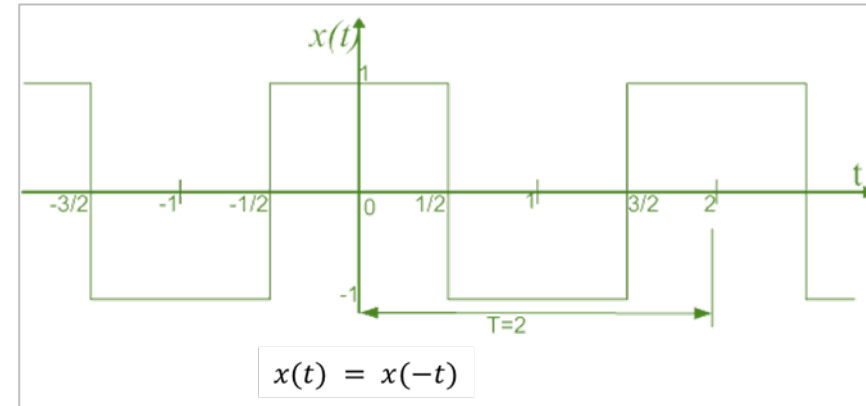
# Fourier Series of an Even Periodic Function

## Even Periodic Function

- A function is defined as **even** if  $x(t) = x(-t)$



Even Periodic Function



- For an **even** periodic function (waveform), the expressions for the Fourier coefficients reduce to

$$A_0 = \frac{2}{T} \int_0^{T/2} x(t) dt$$

$$A_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$$

$$B_n = 0 \text{ for all } n$$

Note that all the 'B' coefficients are zero (sine terms disappear) if the periodic waveform is **even**.

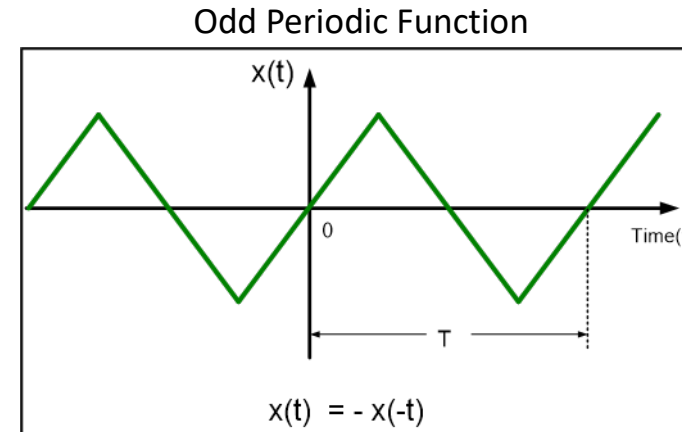
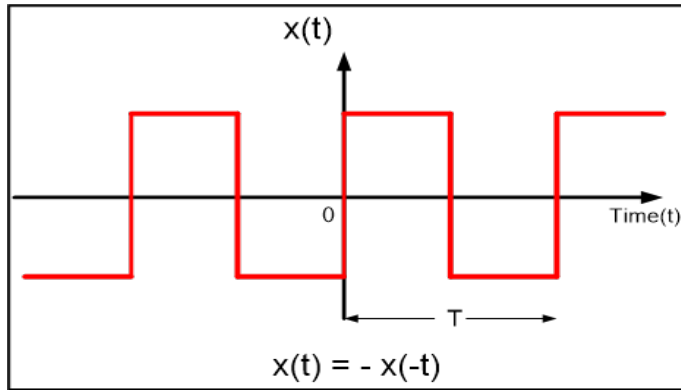
$$\leftarrow x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)$$



# Fourier Series of an Odd Periodic Function

## Odd Periodic Function

- A function is defined as **odd** if  $x(t) = -x(-t)$



- For an **odd** periodic function (waveform), the expressions for the Fourier coefficients reduce to

$$A_0 = 0 \quad \text{and} \quad A_n = 0 \quad \text{for } n > 0$$

$$B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$

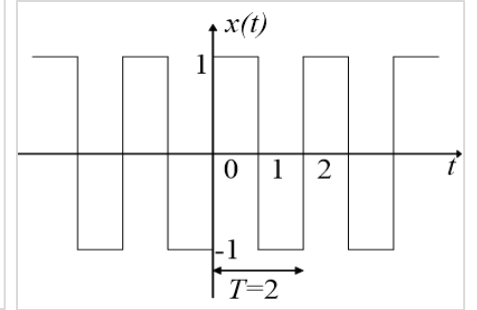
Note that all the 'A' coefficients are zero if the periodic waveform is **odd**. Only odd harmonics Present

# Example 1 (Odd Function)

## Example 1

- Calculate the **Fourier coefficients** for the square wave  $x(t)$  shown on the right. The period  $T$  of the square wave is 2 seconds.

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t) \quad A_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \quad A_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt \quad B_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt$$



## Model Answer

- $\omega_0 = 2\pi/T = 2\pi/2 = \pi$
- Because the above square wave  $x(t)$  is an **odd function**, all the 'A' coefficients are zero: that is,  $A_0 = 0$  and  $A_n = 0$  for all  $n$ . Hence there will be no cosine wave terms in the summation of equation  $x(t)$ .

- The 'B' coefficients are:  $B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$

$$B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt = \frac{4}{2} \int_0^{2/2} 1 \sin(n\pi t) dt = 2 \int_0^1 1 \sin(n\pi t) dt = \frac{2}{n\pi} [1 - \cos(n\pi)]$$

- There will be no cosine wave terms in the summation of equation  $x(t)$  because  $A_n = 0$  for  $n \geq 0$

$$x(t) = \sum_{n=1}^{\infty} \left( \frac{2}{n\pi} (1 - \cos(n\pi)) \right) \sin n\pi t$$



# Example 1 (Odd Function)

$$x(t) = \sum_{n=1}^{\infty} \left( \frac{2}{n\pi} (1 - \cos(n\pi)) \right) \sin n\pi t$$

$$x(t) = \frac{4}{\pi} \sin \pi t + 0 + \frac{4}{3\pi} \sin(3\pi t) + 0 + \frac{4}{5\pi} \sin(5\pi t) + \dots$$

fundamental

2<sup>nd</sup>  
harmonic

4<sup>th</sup>  
harmonic

5<sup>th</sup>  
harmonic

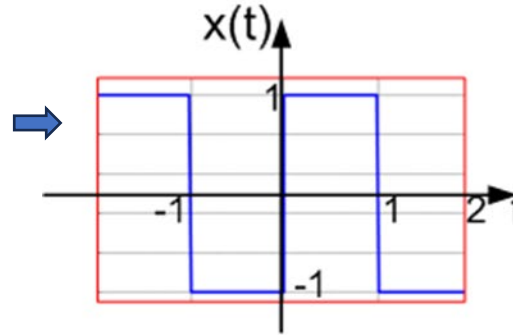
As more harmonics are added the summation gets closer and closer (converges) to  $x(t)$ .

Note:

1. Odd functions can solely be represented by **sine waves** because, sine waves are odd functions. A sum of odd functions is another odd function.
2. We observe that the Fourier series representation of an odd periodic waveform has zero average (DC) value ( $A_0 = 0$ ) and contains only odd harmonics.
3. **Odd function**  $\rightarrow$  only sine terms ( $A_n = 0, A_0 = 0$ )

# How to reconstruct (or synthesise) the signal $x(t)$ from the Fourier coefficients

Original Periodic signal

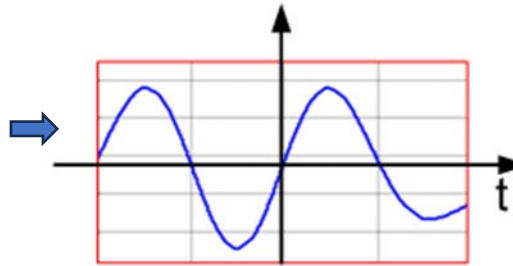


Fourier Series of  $x(t)$ :

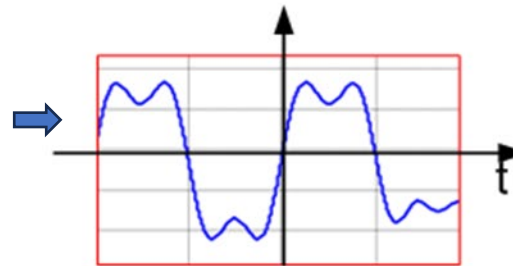
$$x(t) = \frac{4}{\pi} \sin \pi t + 0 + \frac{4}{3\pi} \sin(3\pi t) + 0 + \frac{4}{5\pi} \sin(5\pi t) + \dots$$

Fourier Coefficients

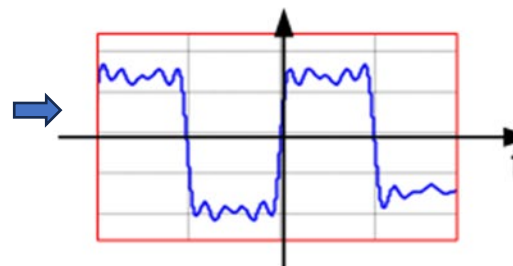
Reconstructed signal with 10 Fourier coefficients



Reconstructed signal with 30 Fourier coefficients



Reconstructed signal with 60 Fourier coefficients



## Note:

1. As more coefficients are used, the reconstructed signal becomes closer to the original signal  $x(t)$
2. In theory, perfect reconstruction is possible using an infinite number of Fourier coefficients.



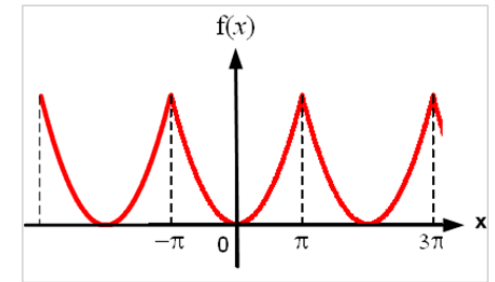
# Example 2 (Even Function)

## Example 2

- Find the Fourier series of the periodic function  $f(x)$  on the right.

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t) \quad A_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \quad A_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt \quad B_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt$$

$$f(x) = x^2 \quad -\pi \leq x \leq \pi$$



## Model Answer

- Period ( $T$ ) =  $2\pi$  rad **Note:**  $x(\text{rad}) = \omega_0 t$
- Because the above function  $f(x)$  is even, all the 'B' coefficients are zero: that is,  $B_n = 0$  for all  $n$ . Hence there will be no sine wave terms in the summation of equation  $f(x)$ .

- The 'A' coefficients are:  $A_0 = \frac{2}{T} \int_0^{T/2} x(t) dt$   $A_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$

$$A_0 = \frac{2}{2\pi} \int_0^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x^2 dx = \frac{\pi^2}{3} \text{ (DC Component)}$$

$$A_n = \frac{4}{2\pi} \int_0^{\pi} f(x) \cos(nx) dx = \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx = \frac{4}{n^2} [\cos(n\pi)]$$

Use integration by parts

## Example 2 (Even Function)

- The Fourier series for  $f(x)$  on the right is:

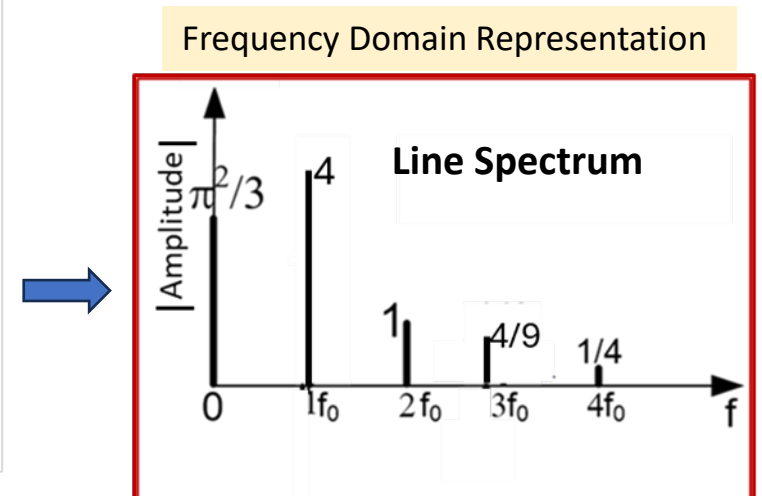
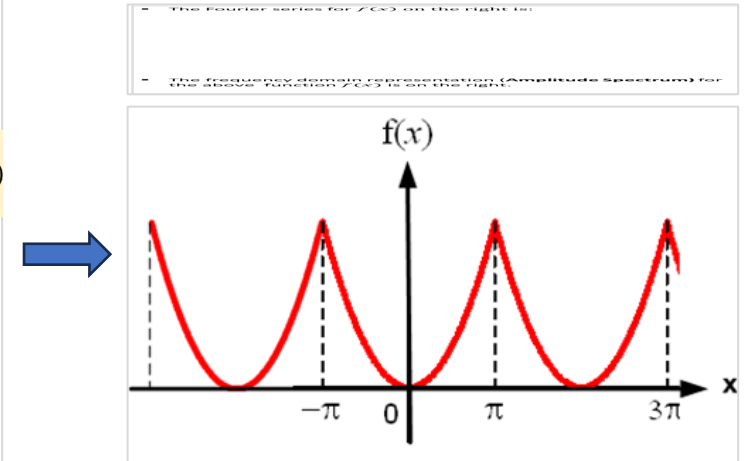
$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left[ \frac{4}{n^2} \cos(n\pi) \right] \cos(nx)$$

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)$$

$$f(x) = \frac{\pi^2}{3} - 4 \left( \cos x - \frac{\cos 2x}{4} + \frac{\cos 3x}{9} - \frac{\cos 4x}{16} + \dots \right)$$

### Note:

- Even functions can solely be represented by cosine waves because, cosine waves are even functions. A sum of even functions is another even function.
  - Even function**  $\rightarrow$  only cosine terms ( $B_n = 0$ )
- The frequency domain representation for the above function  $f(x)$  is on the right. It shows amplitude component of  $f(x)$ .

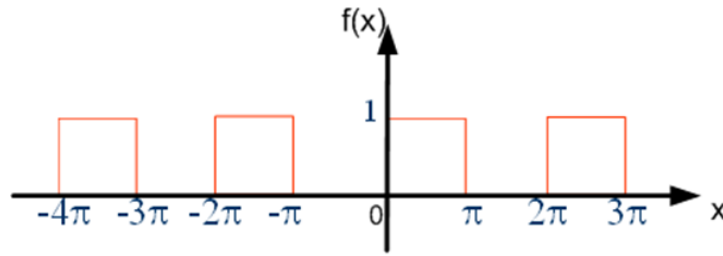




# Example 3 (Neither Even nor Odd)

## Example 3

- Find the Fourier series of the following periodic function (square wave):



## Model Answer

- Period (T) =  $2\pi$  rad    **Note:**  $x(\text{rad}) = \omega_0 t$
- The above function  $f(x)$  is neither even nor odd, so we need to calculate all the Fourier coefficients  $A_0$ ,  $A_n$  and  $B_n$ .
- The 'A' coefficients are:

$$\begin{aligned} A_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx \\ &= \frac{1}{2\pi} \int_{-\pi}^0 0 dx + \int_0^{\pi} 1 dx = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} A_n &= \frac{2}{2\pi} \int_0^{\pi} \cos(nx) dx \\ &= \frac{1}{n\pi} [\sin(nx)]_0^{\pi} = 0 \end{aligned}$$

$$n=1, 2, 3, \dots$$

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)$$

$$A_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt$$

$$A_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt$$

$$B_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt$$

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T} \text{ radians/sec.}$$

# Example 3

## Model Answer

- The 'B' coefficients are:

$$\begin{aligned} B_n &= \frac{2}{2\pi} \int_0^\pi \sin(nx) dx \\ &= -\frac{1}{n\pi} (\cos n\pi - 1) \\ &= \begin{cases} 2/n\pi & n = 1, 3, 5, \dots \\ 0 & n = 2, 4, 6, \dots \end{cases} \end{aligned}$$

- The corresponding Fourier series is:

$$f(x) = \frac{1}{2} + \frac{2}{\pi} \left( \sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right)$$

**Note:** Even harmonic components are zero.

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)$$

$$\begin{aligned} A_0 &= 1/2 \\ A_n &= 0 \end{aligned}$$

$$B_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt$$





# ELEC2134 : Circuits and Signals (PART A)

## Fourier series: (B) An Alternative Trigonometric Form

- In the frequency domain, the amplitude of each frequency component is determined by  $A_n$  and  $B_n$ , with contributions from both the sine and cosine terms  $\{x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)\}$ .
- We would like to express the signal in an alternative form in which the amplitude of each frequency component is obtained using a single cosine function with a phase angle:  $\{x(t) = K_0 + \sum_{n=1}^{\infty} K_n \cos(n\omega_0 t + \phi_n)\}$

# Fourier Series: (B) An Alternative Trigonometric Form

- The trigonometric Fourier series is given by:  $x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)$
- In many applications of the Fourier series, the cosine and sine terms are combined into a single term (either a cosine or a sine expression) for convenience. An alternative form of  $x(t)$  is the **amplitude – phase** representation:  $x(t) = K_0 + \sum_{n=1}^{\infty} K_n \cos(n\omega_0 t + \phi_n)$



$$x(t) = K_0 + K_1 \cos(1\omega_0 t + \phi_1) + K_2 \cos(2\omega_0 t + \phi_2) + K_3 \cos(3\omega_0 t + \phi_3) + \dots$$

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t) \rightarrow x(t) = K_0 + \sum_{n=1}^{\infty} K_n \cos(n\omega_0 t + \phi_n)$$

$$\{ K_n \cos(n\omega_0 t + \phi_n) = \{ K_n \cos(\phi_n) \cos(n\omega_0 t) - (K_n \sin(\phi_n)) \sin(n\omega_0 t) \} \quad A_n = K_n \cos(\phi_n); \quad B_n = -K_n \sin(\phi_n)$$

$$\text{DC Component } (K_0) = A_0; \quad \text{Harmonic Amplitude } (K_n) = \sqrt{A_n^2 + B_n^2};$$

$$\text{Phase Angle } (\phi_n) = \tan^{-1} \left( \frac{-B_n}{A_n} \right) = -\tan^{-1} \left( \frac{B_n}{A_n} \right); \quad \text{Also } K_n \angle \phi_n = A_n - jB_n$$



# Example 1 (Amplitude – Phase Form)

- Find the Fourier Coefficients  $A_0$ ,  $A_n$  and  $B_n$  as well as the amplitude-Phase Fourier series for the sawtooth waveform (**Neither Even nor Odd**) given below:

$$A_0 = \frac{1}{T} \int_0^T \frac{t}{T} dt = \frac{1}{2};$$

$$A_n = \frac{2}{T} \int_0^T \frac{t}{T} \cos(n\omega_0 t) dt = \frac{2}{T^2} \int_0^T t \cos(n\omega_0 t) dt = \frac{2}{T^2} \left\{ \left[ \frac{\cos(n\omega_0 t)}{(n\omega_0)^2} + \frac{t \sin(n\omega_0 t)}{n\omega_0} \right]_0^T \right\} \text{ (using integration by parts)}$$

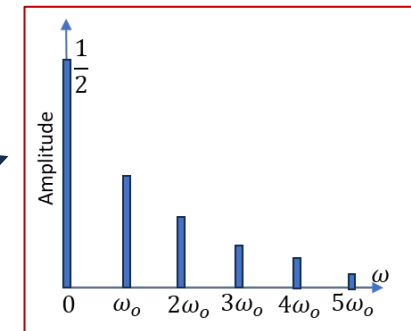
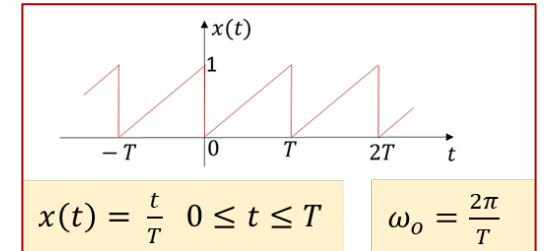
$$A_n = \frac{2}{T^2} \left\{ \frac{\cos(2n\pi) - 1}{(n\omega_0)^2} + \frac{T \sin(2n\pi) - 0}{n\omega_0} \right\} = 0 \text{ (i.e., the cosine terms are missing for this particular waveform)}$$

$$B_n = \frac{2}{T} \int_0^T \frac{t}{T} \sin(n\omega_0 t) dt = \frac{2}{T^2} \int_0^T t \sin(n\omega_0 t) dt = \frac{2}{T^2} \left\{ \left[ \frac{\sin(n\omega_0 t)}{(n\omega_0)^2} - \frac{t \cos(n\omega_0 t)}{n\omega_0} \right]_0^T \right\} \text{ (using integration by parts)}$$

$$B_n = \frac{2}{T^2} \left\{ \frac{\sin(2n\pi) - 0}{(n\omega_0)^2} + \frac{T \cos(2n\pi) - 0}{n\omega_0} \right\} = \frac{2}{T^2} \left( \frac{-T}{n\omega_0} \right) = -\frac{1}{n\pi}$$

The **Fourier series** is:  $x(t) = \frac{1}{2} - \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin(n\omega_0 t)$

The **Amplitude – Phase Series** is:  $x(t) = \frac{1}{2} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \cos(n\omega_0 t + 90^\circ)$



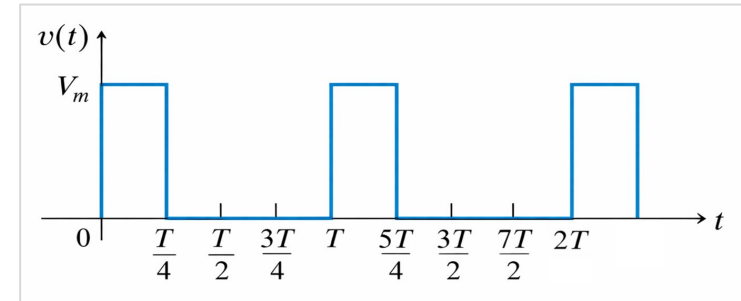
For a sawtooth wave, the amplitude of each harmonic decreases as  $1/n$ , and all harmonics have the same phase.



# Example 2 (Amplitude – Phase Form)

## Example 2

- (a) Derive the expressions for  $A_n$  and  $B_n$  for the periodic function shown on the right:
- (b) Write the first four terms of the Fourier series representation of  $v(t)$  using the format of  $x(t) = K_0 + \sum_{n=1}^{\infty} K_n \cos(n\omega_0 t + \phi_n)$  {Note:  $K_n \angle \phi_n = A_n - jB_n$ }



## Model Answer

- The waveform  $v(t)$  has amplitude  $V_m$ , period  $T$  and is nonzero only over the interval  $0 \leq t \leq T/4$  in each period. The voltage  $v(t)$  is neither **even** nor **odd**.

$$(a) \quad A_n = \frac{2}{T} \left[ \int_0^{T/4} V_m \cos(n\omega_0 t) dt + \int_{T/4}^T 0 \cos(n\omega_0 t) dt \right]$$

$$A_n = \frac{2V_m}{T} \left[ \frac{\sin(n\omega_0 t)}{n\omega_0} \right]_0^{T/4}; \quad A_n = \frac{V_m}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

$$B_n = \frac{2}{T} \left[ \int_0^{T/4} V_m \sin(n\omega_0 t) dt \right]$$

$$B_n = \frac{2V_m}{T} \left[ \frac{-\cos(n\omega_0 t)}{n\omega_0} \right]_0^{T/4}; \quad B_n = \frac{V_m}{n\pi} \left[ 1 - \cos\left(\frac{n\pi}{2}\right) \right]$$

$$(b) \quad \text{The average value of } v(t) \text{ is: } A_0 = \frac{V_m \left(\frac{T}{4}\right)}{T} = \frac{V_m}{4}$$

The values of  $A_n - jB_n$  for  $n = 1, 2$  and  $3$  are:

$$A_1 - jB_1 = \frac{V_m}{\pi} - j \frac{V_m}{\pi} = \frac{\sqrt{2}V_m}{\pi} \angle -45^\circ$$

$$A_2 - jB_2 = 0 - j \frac{V_m}{\pi} = \frac{V_m}{\pi} \angle -90^\circ$$

$$A_3 - jB_3 = -\frac{V_m}{3\pi} - j \frac{V_m}{3\pi} = \frac{\sqrt{2}V_m}{3\pi} \angle -135^\circ$$

The first four terms in the Fourier series representation of  $v(t)$  are:

$$v(t) = \frac{V_m}{4} + \frac{\sqrt{2}V_m}{\pi} \cos(\omega_0 t - 45^\circ) + \frac{V_m}{\pi} \cos(2\omega_0 t - 90^\circ) + \frac{\sqrt{2}V_m}{3\pi} \cos(3\omega_0 t - 135^\circ) + \dots$$

# ELEC2134 : Circuits and Signals (PART A)

## Fourier Series: (C) Exponential Fourier series

- A more convenient and compact representation of the Fourier series is obtained by replacing the sine and cosine terms with complex exponential functions, which combine the amplitude and phase into a single coefficient  $C_n$ .  $\{x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}\}$

{ Instead  $x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)$  }

- Each coefficient  $C_n$  contains:
  - Magnitude  $|C_n| \rightarrow$  amplitude of the harmonic
  - Angle  $\angle C_n \rightarrow$  phase of the harmonic(Amplitude and phase are combined into a single coefficient  $C_n$ )

# Complex Form of the Fourier Series (Exponential Fourier Series)

- The cosine–sine and amplitude–phase Fourier series that we have studied so far are referred to as the trigonometric Fourier series.
- An alternative Fourier representation uses exponential functions instead of trigonometric functions. This form is therefore called the exponential Fourier series and is given by:

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}; \quad C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt \quad \text{where } \omega_0 = \frac{2\pi}{T}$$

$$\text{DC Component } A_0 = C_0; \quad C_n = \frac{1}{2}(A_n - jB_n); \quad |C_{-n}| = |C_n| = \frac{1}{2}\sqrt{A_n^2 + B_n^2}$$

- For Even Symmetry:

$$C_n = \frac{2}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$$

- For Odd Symmetry:

$$C_n = \frac{-2j}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$



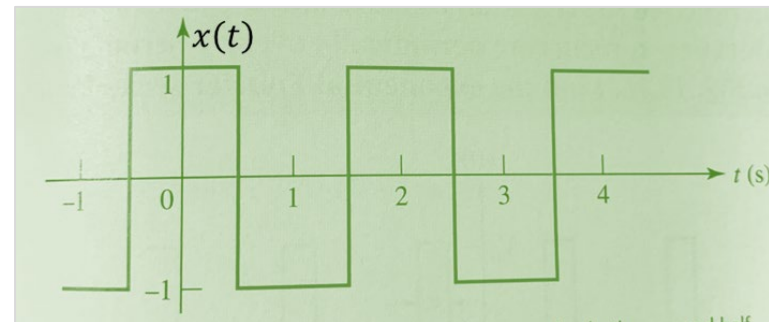
# Exponential Fourier Series: Example 1

**Example 1:** Determine  $C_n$  for the square wave shown on the right.

## Model Answer

- This square wave possesses both **even symmetry**. If we ignore the symmetry and use the exponential Fourier series equation, with  $T=2$ ,  
 $\therefore \omega_0 = \frac{2\pi}{T} = \pi$

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt \quad \text{or} \quad C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$



$$\begin{aligned} C_n &= \frac{1}{2} \left[ \int_{-1}^{-0.5} (-1) e^{-jn\pi t} dt + \int_{-0.5}^{0.5} e^{-jn\pi t} dt - \int_{0.5}^1 e^{-jn\pi t} dt \right] \\ &= \frac{1}{2} \left[ \frac{-1}{-jn\pi} (e^{-jn\pi t})_{-1}^{-0.5} + \frac{1}{-jn\pi} (e^{-jn\pi t})_{-0.5}^{0.5} + \frac{-1}{-jn\pi} (e^{-jn\pi t})_{0.5}^1 \right] \\ &= \frac{1}{j2n\pi} \left( e^{jn\pi/2} - e^{jn\pi} - e^{-jn\pi/2} + e^{jn\pi/2} + e^{-jn\pi} - e^{-jn\pi/2} \right) \\ &= \frac{1}{j2n\pi} \left( 2e^{jn\pi/2} - 2e^{-jn\pi/2} \right) \\ &= \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right). \end{aligned} \quad \therefore x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

- Using Symmetry (even symmetry), there is less work when we apply the corresponding symmetry equations

$$\begin{aligned} C_n &= \frac{4}{T} \int_0^{T/4} f(t) \cos(n\omega_0 t) dt \\ &= \frac{4}{2} \int_0^{0.5} \cos(n\pi t) dt \\ &= \frac{2}{n\pi} [\sin(n\pi t)]_0^{0.5} \\ &= \begin{cases} \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right), & n \text{ odd} \\ 0, & n \text{ even} \end{cases} \end{aligned}$$

From this result,  $c_0 = 0$ ,  $c_1 = \frac{2}{\pi}$ ,  $c_2 = 0$ ,  $c_3 = -\frac{2}{3\pi}$ ,  $c_4 = 0$ ,  $c_5 = \frac{2}{5\pi}$ , and so on.



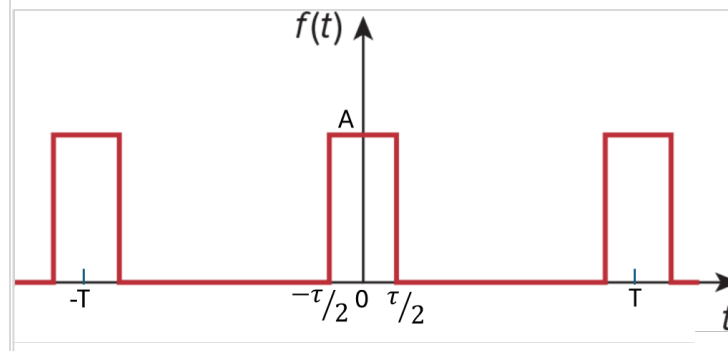
# Exponential Fourier Series: Example 2

**Example 2:** A periodic rectangular pulse train  $f(t)$  has period  $T = 10\text{s}$  and amplitude  $A = 10$ . In each period, the pulse is centred at  $t = 0$  and has pulse width  $\tau$ (seconds), as shown on the right.

(a) Find the **exponential Fourier series** coefficients  $C_n$  in

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T}.$$

(b) Sketch the **line spectrum**  $|C_n|$  versus harmonic number  $n$  for three pulse widths:  
 $\tau = 2, 4, 6\text{ s}$ .



## Model Answer

**Exponential Fourier series coefficient**

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt.$$

Since  $f(t) = A$  between  $-\frac{\tau}{2}$  and  $\frac{\tau}{2}$

$$C_n = \frac{1}{T} \int_{-\tau/2}^{\tau/2} A e^{-jn\omega_0 t} dt = \frac{A}{T} \int_{-\tau/2}^{\tau/2} e^{-jn\omega_0 t} dt.$$

▪ DC Term ( $n = 0$ ):  $C_0 = \frac{A}{T} \int_{-\tau/2}^{\tau/2} 1 dt = \frac{A\tau}{T}$

$$\text{For } n \neq 0 \quad C_n = \frac{1}{T} \int_{-\tau/2}^{\tau/2} A e^{-jn\omega_0 t} dt$$

$$\begin{aligned} C_n &= \frac{A}{T} \left[ \frac{e^{-jn\omega_0 t}}{-jn\omega_0} \right]_{-\tau/2}^{\tau/2} = \frac{A}{T} \cdot \frac{1}{-jn\omega_0} (e^{-jn\omega_0 \tau/2} - e^{+jn\omega_0 \tau/2}) \\ &= \frac{A}{T} \cdot \frac{1}{-jn\omega_0} (-2j \sin(\frac{n\omega_0 \tau}{2})) = \frac{A}{T} \cdot \frac{2 \sin(\frac{n\omega_0 \tau}{2})}{n\omega_0}. \end{aligned}$$



# Exponential Fourier Series: Example 2

Using  $\omega_0 = \frac{2\pi}{T}$ , this simplifies to:

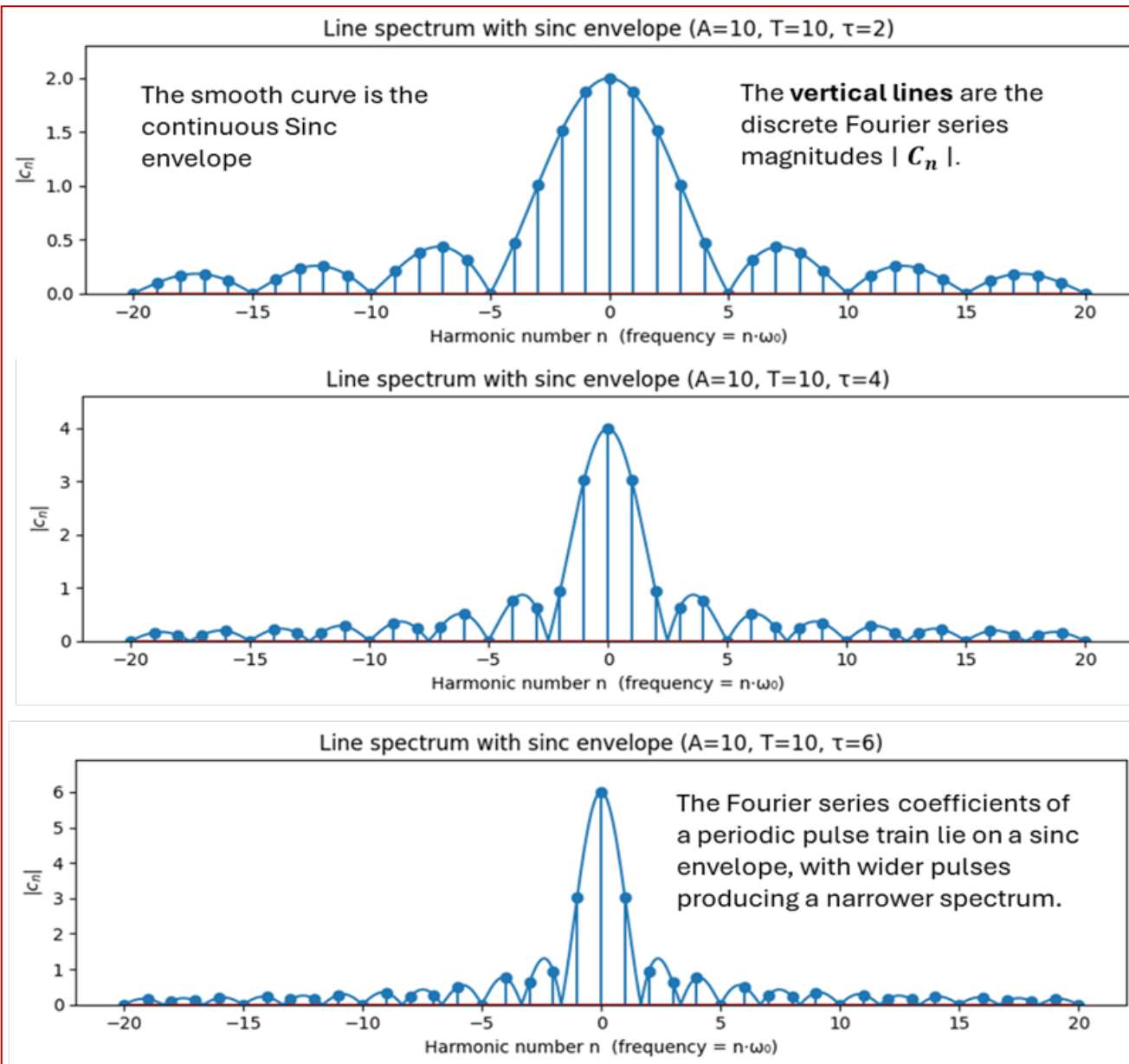
$$C_n = \begin{cases} \frac{A\tau}{T}, & n = 0 \\ \frac{A}{n\pi} \sin\left(\frac{n\pi\tau}{T}\right), & n \neq 0 \end{cases}$$

When  $A=10$ ,  $T=10$ ,

$$C_0 = \tau, \quad C_n = \frac{10}{n\pi} \sin\left(\frac{n\pi\tau}{10}\right), \quad n \neq 0$$

## (b) Line spectrum:

- A **line spectrum** plots  $|C_n|$  at discrete frequencies  $n\omega_0$ . The line spectra  $|C_n|$  for  $\tau = 2, 4, 6$ s (with  $A = 10$ ,  $T = 10$ ) is shown on the right.
- The envelope of  $|C_n|$  follows a sinc-like shape: wider pulses  $\rightarrow$  **narrower** spectrum (more energy concentrated near  $n = 0$ ).



# ELEC2134 : Circuits and Signals (PART A)

## Half-Wave Symmetry

- In Fourier series analysis, half-wave symmetry (The waveform repeats every  $\frac{T}{2}$  with a sign change) is useful because it eliminates entire sets of harmonics, making both the analysis and computation much simpler.
- For a signal with half-wave symmetry:  $A_n = B_n = 0$  for even  $n$  and only odd harmonics remain.

# Fourier Series: Half-Wave Symmetry and Odd Half-Wave Symmetry

- Another important Type of symmetry is the **half-wave Symmetry**. A periodic function possesses half-wave symmetry if

$$x(t) = -x(t - \frac{T}{2})$$

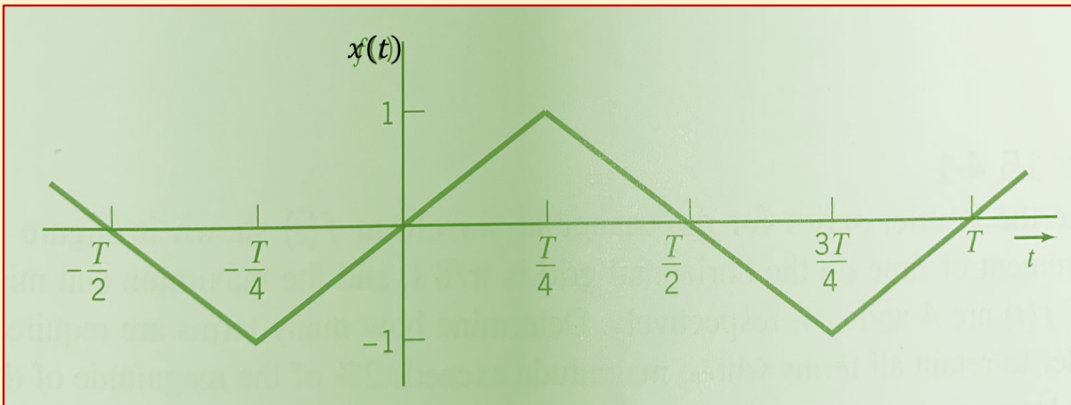
- Shifting such a function by half period and then inverting it yields again the original function.
- If a function has half-wave symmetry, then both  $A_n$  and  $B_n$  are zero for **even** values of  $n$ .

For a function with a **half-wave** symmetry:

$$A_n = \begin{cases} \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even} \end{cases}$$

$$B_n = \begin{cases} \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even} \end{cases}$$

- The diagram shown below is an **odd** function ( $A_0=0$ ) with **half-wave** symmetry



A periodic signal  $x(t)$  has **odd half-wave symmetry** if **both** of the following are true:

$$x(t) = -x(t - \frac{T}{2}) \quad \text{and} \quad x(t) = -x(-t)$$

For **odd half-wave** symmetry:

- All cosine terms are zero  $A_n = 0$
- All even harmonics are zero
- Only odd sine harmonics appear in the Fourier series

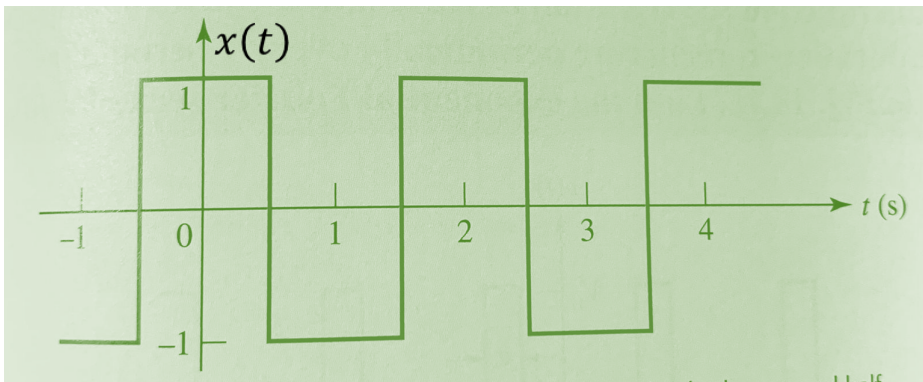


# Fourier Series: Even Half-Wave Symmetry

A periodic signal  $x(t)$  has even **half-wave symmetry** if **both** of the following are true:

$$x(t) = -x\left(t - \frac{T}{2}\right) \quad \text{and} \quad x(t) = x(-t)$$

- An **even and half wave symmetry** waveform is shown below:



If a waveform has **even half-wave symmetry**, then:

- All sine coefficients  $B_n = 0$
- All even harmonics are zero
- Only odd cosine harmonics remain

For a function with a **half-wave** symmetry:

$$A_n = \begin{cases} \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even} \end{cases}$$



$$B_n = \begin{cases} \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even} \end{cases}$$

Note:

1. Even half-wave symmetry exists only when a waveform is both even and has half-wave symmetry, which eliminates sine terms and even harmonics in its Fourier series.
2. Odd half-wave symmetry exists only when a waveform is both odd and has half-wave symmetry, which eliminates cosine terms and even harmonics in its Fourier series.



# Example 1 (Half-Wave Symmetry)

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)$$

**Example 1:** The periodic signal  $x(t)$  shown in the figure has period  $T = 2$ s.

- Determine whether the signal has odd symmetry and/or half-wave symmetry.
- Using symmetry arguments, identify which Fourier coefficients are zero.
- Derive the trigonometric Fourier series of  $x(t)$ .
- Write the final Fourier series in simplified form.

## Model Answer

(a) The waveform shown is an odd function, i.e.  $x(t) = -x(-t)$ . It also satisfies  $x(t) = -x(t - T/2)$ . Hence, the signal has **half-wave symmetry** (see the figure on the right). **Note:**  $\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$

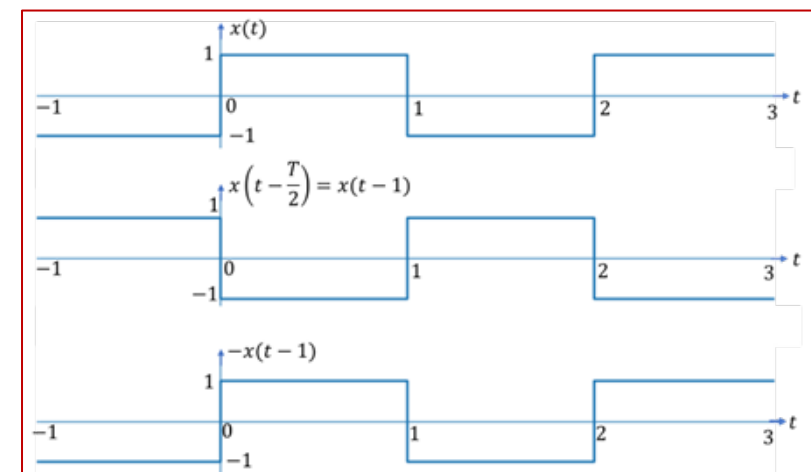
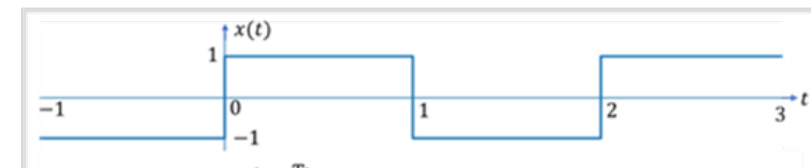
(b) From symmetry:

- Odd symmetry**  $\Rightarrow A_0 = 0$  and  $A_n = 0$
- Half-wave symmetry**  $\Rightarrow B_n = 0$  for even  $n$

$$(c) \quad B_n = \frac{2}{T} \int_0^T x(t) \sin(n\pi t) dt = \int_{-1}^1 x(t) \sin(n\pi t) dt = \int_{-1}^0 (-1) \sin(n\pi t) dt + \int_0^1 (1) \sin(n\pi t) dt$$

$$B_n = \frac{2(1 - \cos(n\pi))}{n\pi}; \text{ Since } \cos(n\pi) = (-1)^n, B_n = \begin{cases} \frac{4}{n\pi} & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

$$(d) \text{ Final Fourier Series: } x(t) = \frac{4}{\pi} \left[ \sin(\pi t) + \frac{1}{3} \sin(3\pi t) + \frac{1}{5} \sin(5\pi t) + \dots \right] = \sum_{n=1 \text{ \& } n \text{ odd}}^{\infty} \frac{4}{n\pi} \sin(n\pi t)$$



Odd and half-wave symmetry eliminate the DC term, all cosine terms, and all even harmonics—leaving only odd sine harmonics.

There are three types of symmetry:

(1) even-function symmetry

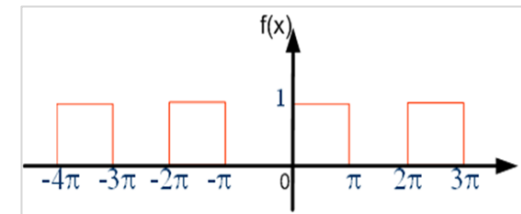
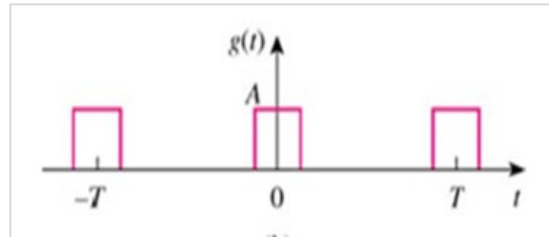
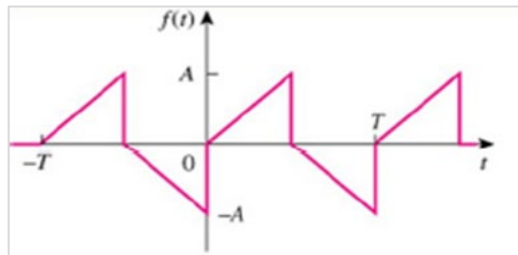
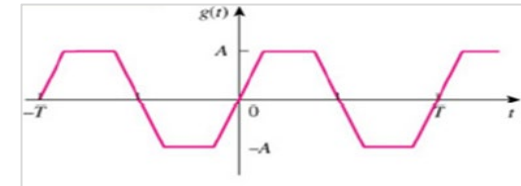
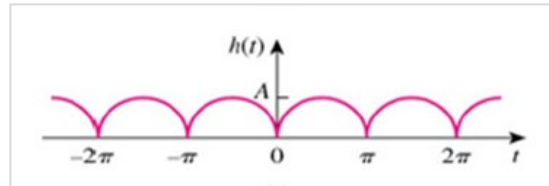
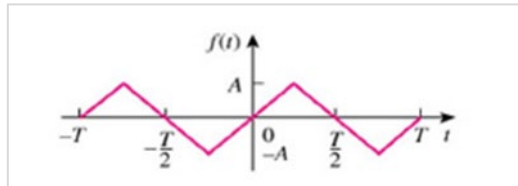
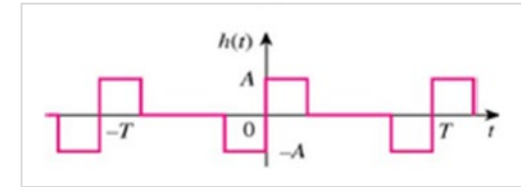
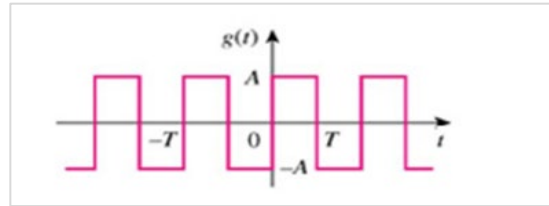
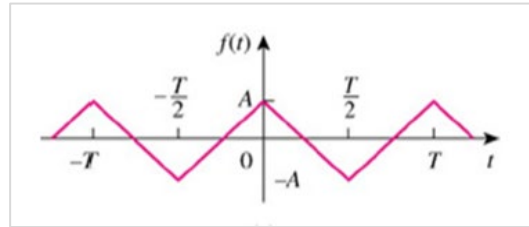
(2) odd-function symmetry

(3) half-wave symmetry

$$f(t) = f(-t)$$

$$f(t) = -f(-t)$$

$$f(t) = -f(t - T/2)$$



# Fourier Series: Summary

$$A_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt$$

$$A_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt$$

$$B_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt$$

(A)  $x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t);$

(B)  $x(t) = K_0 + \sum_{n=1}^{\infty} K_n \cos(n\omega_0 t + \phi_n); \quad K_n \angle \phi_n = A_n - jB_n; \quad K_n = \sqrt{A_n^2 + B_n^2}; \quad \phi_n = -\tan^{-1}\left(\frac{B_n}{A_n}\right)$

## Odd function

$$x(t) = -x(-t)$$

$$A_n = 0 \text{ for all } n$$

$$B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$

## Even function

$$x(t) = x(-t)$$

$$B_n = 0 \text{ for all } n$$

$$A_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$$

## Half-wave symmetry

$$x(t) = -x\left(t - \frac{T}{2}\right)$$

$$A_n = 0 \text{ for even } n; \quad B_n = 0 \text{ for even } n$$

$$A_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \text{ for odd } n$$

$$B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \text{ for odd } n$$

## (C) Exponential Fourier Series

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}; \quad C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt; \quad C_n = \frac{1}{2}(A_n - jB_n)$$

- For Even Symmetry:

$$C_n = \frac{2}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$$

- For Odd Symmetry:

$$C_n = \frac{-2j}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$

### For odd half-wave symmetry:

- All cosine terms are zero  $A_n = 0$
- All even harmonics are zero
- Only odd sine harmonics remain

### For even half-wave symmetry:

- All sine coefficients  $B_n = 0$
- All even harmonics are zero
- Only odd cosine harmonics remain

# ELEC2134: Circuits and Signals

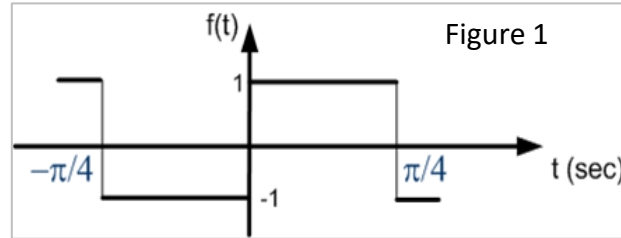
## Transform Methods : Fourier Series

Tutorial questions

# Pass-Level Tutorial Questions (Fourier Series)

**Question 1:** Determine the following for the waveform  $f(t)$  shown in figure 1:

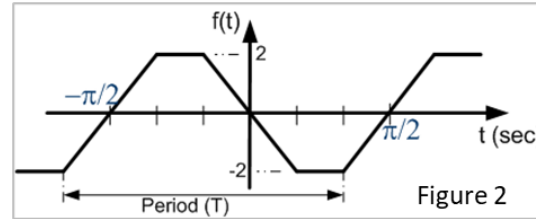
- (a) the period  $T$ ;
- (b) the fundamental frequency  $\omega_o$  ;
- (c) the Fourier Series representation of  $f(t)$ .



**Ans:** (a)  $T = \frac{\pi}{2}$  (sec) (b)  $\omega_o = 4$  rad/s

(c) 
$$f(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin(4nt)$$

**Question 2:** Determine the Fourier series for the waveform  $f(t)$  shown in Figure 2.



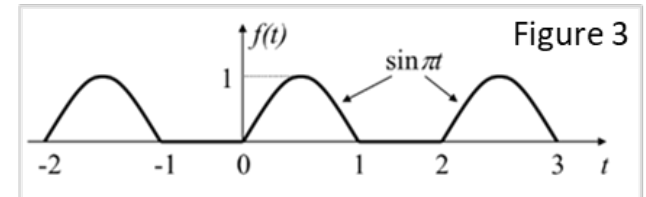
**Ans:** 
$$f(t) = \frac{-24}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin\left(\frac{n\pi}{3}\right) \sin(2nt)$$

**Question 3:** A half-wave rectified sine wave is shown in Figure 3.

- (a) Show that  $f(t)$  can be expressed using Fourier components as:

$$f(t) = \frac{1}{\pi} + \frac{1}{2} \sin \pi t + \sum_{n=2}^{\infty} \frac{1 - \cos(1+n)\pi}{\pi(1-n^2)} \cos n\pi t$$

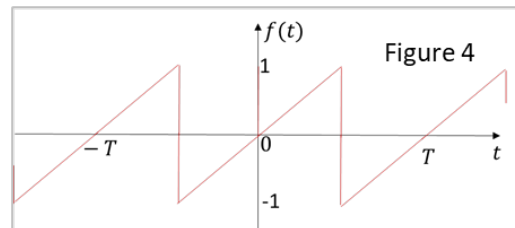
- (b) Using the results obtained in part (a), show that the 4<sup>th</sup> harmonic component of  $f(t)$  is given by  $-\frac{2}{15\pi}$ .



**Question 4:** A sawtooth waveform  $f(t)$  is shown in Figure 4.

Show that  $f(t)$  can be expressed using Fourier components as:

$$f(t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sin(n\omega_0 t)$$







# Pass-Level Tutorial Questions (Fourier Series)

**Question 5:** Sketch each of the functions described, state whether or not even symmetry, odd symmetry, and half-wave symmetry are present, and give the period:

(a)  $v = 0$ ,  $-2 < t < 0$  and  $2 < t < 4$   
 $v = 5$   $0 < t < 2$   
 $v = -5$   $4 < t < 6$   
 and the waveform repeats.

(b)  $v = 10$   $1 < t < 3$   
 $v = 0$   $3 < t < 7$   
 $v = -10$   $7 < t < 9$   
 and the waveform repeats.

c)  $v = 8t$   $-1 < t < 1$   
 $v = 0$   $1 < t < 3$   
 and the waveform repeats.

Ans:

(a) no, no, yes; Period = 8.

(b) no, no, no; Period = 8.

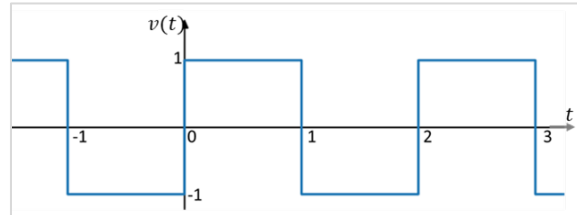
(c) no, yes, no; Period = 4.

(d) Determine the Fourier series for the waveform (a) above:

Ans:

$$\sum_{n=1(\text{odd})}^{\infty} \frac{10}{n\pi} \left( \sin \frac{n\pi}{2} \cos \frac{n\pi t}{4} + \sin \frac{n\pi t}{4} \right);$$

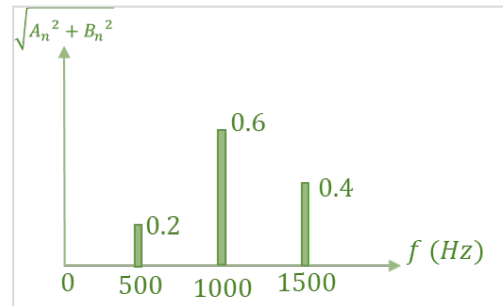
**Question 6:** Determine the general coefficient  $C_n$  in the complex Fourier series for the waveform shown below:



Ans:

$$v(t) = \frac{-j2}{n\pi} \text{ for } n \text{ odd, } 0 \text{ for } n \text{ even}$$

**Question 7:** The periodic function  $f(t)$  is known to have odd symmetry and the amplitude spectrum is shown below. If all coefficients  $A_n$  and  $B_n$  are nonnegative, determine the Fourier series  $f(t)$ .



Ans:

$$y(t) = 0.2 \sin(1000\pi t) + 0.6 \sin(2000\pi t) + 0.4 \sin(3000\pi t)$$





# Higher-Level Tutorial Questions (Fourier Series)

**Question 1:** Given  $x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)$ , Show that the Fourier coefficients  $A_n$  and  $B_n$  are given by the following equations:

$$A_n = \frac{2}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \quad B_n = \frac{2}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$

**Question 2:** Using the trigonometric Fourier series for  $x(t)$  and the Fourier coefficient equations given in Question 1 above, show that the exponential Fourier series can be written in the form shown below. Note that the coefficients  $C_n$  are generally complex numbers.

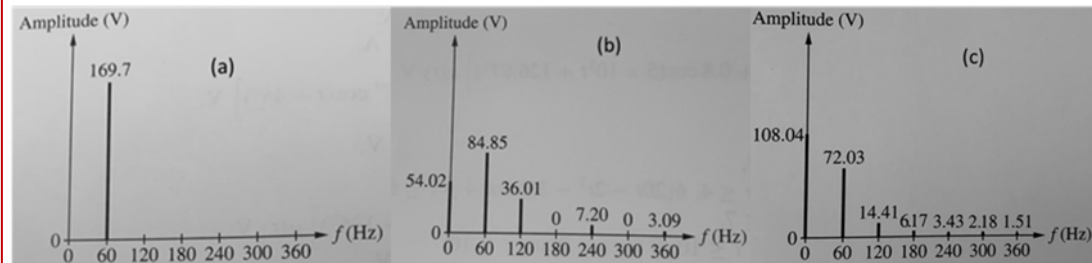
$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}; \quad C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

**Question 3:** By finding the Fourier coefficients for the following parts, sketch and label the amplitude spectrum of each waveform.

- (a)  $v(t) = 120\sqrt{2} \sin(2\pi 60t)$  Volts
- (b) the waveform obtained by half-wave rectifying  $v(t)$
- (c) the waveform obtained by full-wave rectifying  $v(t)$ .

Compare the spectra of parts (a), (b) and (c) above, and comment on the unique features of each.

Ans:



**Question 4:** A signal with fundamental period  $T$  is said to possess half-wave symmetry if it satisfies

$$x(t) = -x\left(t - \frac{T}{2}\right).$$

That is, half of one period of the signal is the negative of the other half.

Show that the Fourier coefficients associated with even harmonics, are zero for all signals with half-wave symmetry.

# ELEC2134: Circuits and Signals

## Appendix A: Filters

# Filters

Following are the idealised specifications of the most filter functions:

## Ideal Low-Pass Filter

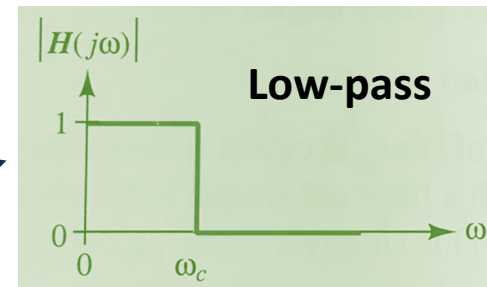
$$|H(j\omega)| = \begin{cases} 1, & 0 \leq \omega < \omega_c \\ 0, & \omega > \omega_c \end{cases}$$

Passband:  $0 \leq \omega < \omega_c$

Stopband:  $\omega_c < \omega \leq \infty$

### Low-pass:

Passes low frequencies,  
blocks high frequencies.

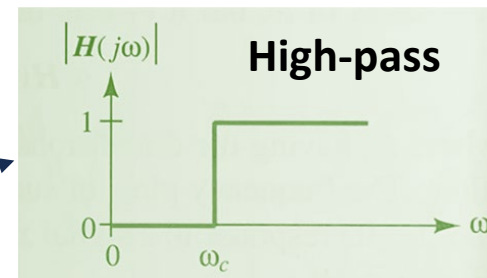


## Ideal High-Pass Filter

$$|H(j\omega)| = \begin{cases} 0, & 0 \leq \omega < \omega_c \\ 1, & \omega > \omega_c \end{cases}$$

### High-pass:

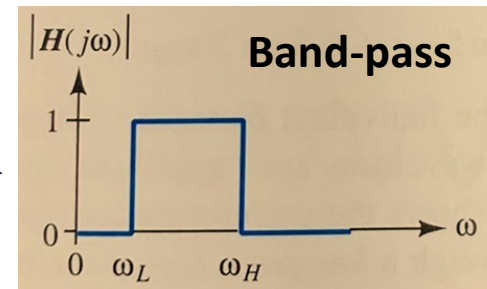
Passes high frequencies,  
blocks low frequencies.



## Ideal Band-Pass Filter

$$|H(j\omega)| = \begin{cases} 1, & \omega_L \leq \omega \leq \omega_H \\ 0, & \text{otherwise} \end{cases}$$

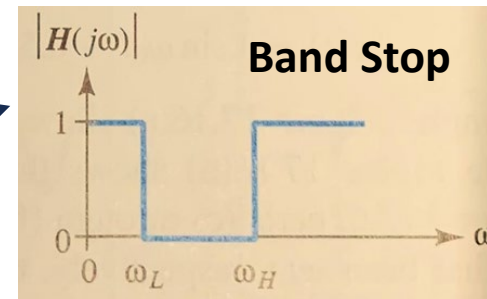
**Band-pass:** Passes only  
frequencies between  
 $\omega_L$  and  $\omega_H$ .



## Ideal Band-Stop (Band-Reject) Filter

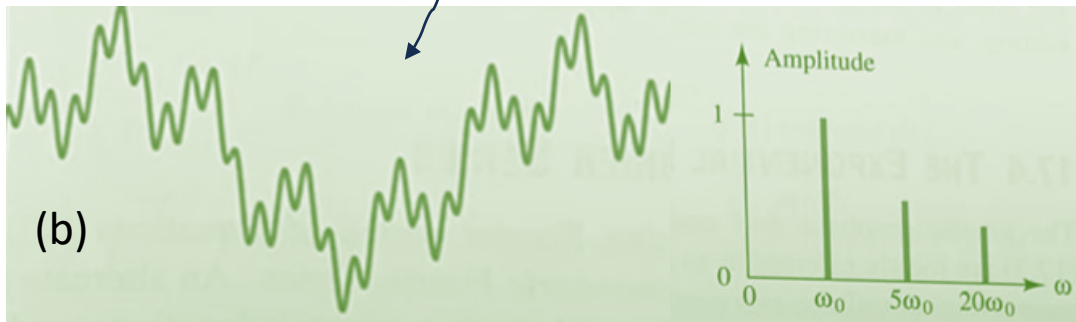
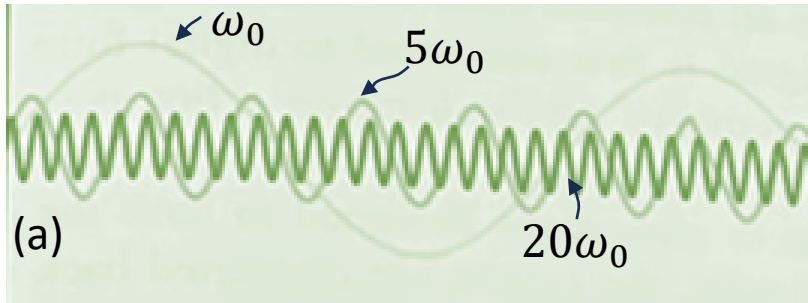
$$|H(j\omega)| = \begin{cases} 0, & \omega_L \leq \omega \leq \omega_H \\ 1, & \text{otherwise} \end{cases}$$

**Band-stop:** Rejects  
frequencies between  
 $\omega_L$  and  $\omega_H$ .

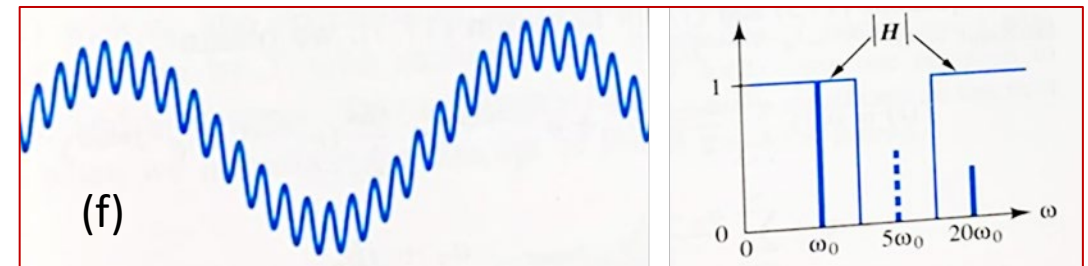
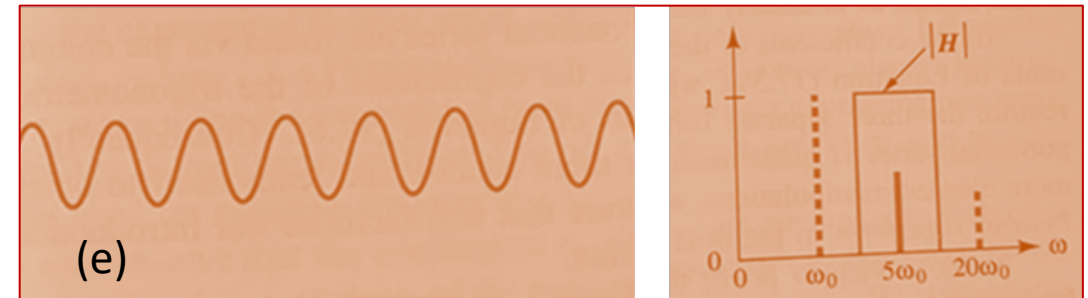
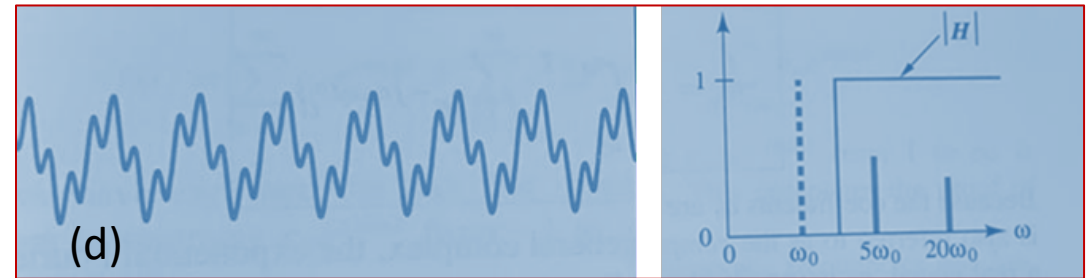
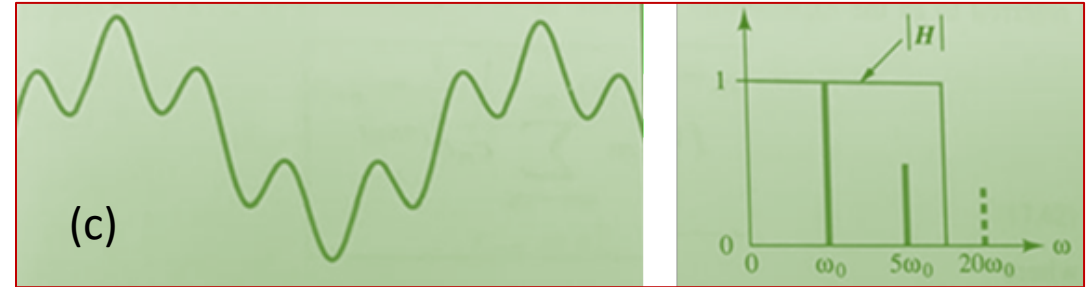


# Filters

- The effect of different filters is illustrated using  $x(t) = \sin(\omega_0 t) + 0.5\sin(5\omega_0 t) + 0.3\sin(20\omega_0 t)$ .
- Figure (a) shows the harmonic components.
- Figure (b) shows the waveform and spectrum.
- Figures (c) to (f) show the outputs of low-pass, high-pass, band-pass, and band-reject filters.



$$x(t) = \sin(\omega_0 t) + 0.5\sin(5\omega_0 t) + 0.3\sin(20\omega_0 t).$$

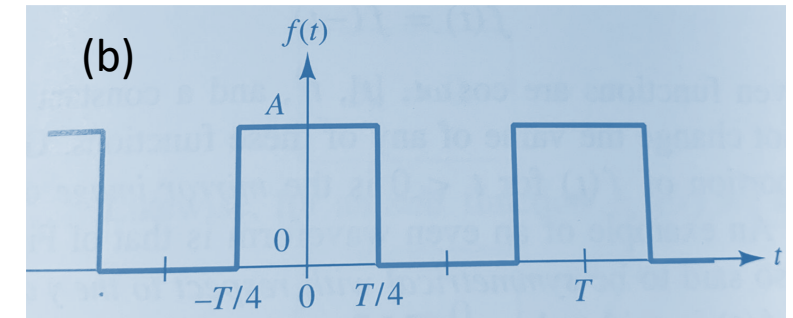
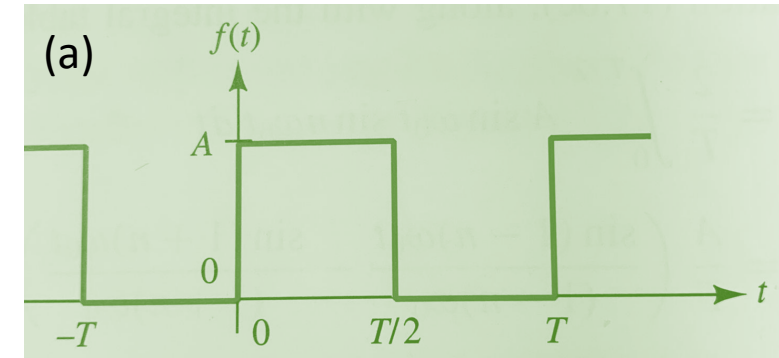


# Effect of Shifting

- Shifting a periodic function  $f(t)$  along either the vertical or the horizontal axis will generally affect its Fourier coefficients.
- The coefficients are also affected by the presence of certain types of functional symmetries in  $f(t)$ .
- Of particular interest are even-function, odd-function, and half-period symmetry because some coefficient sets vanish altogether, expediting our calculations.

## Shifting

- Shifting a periodic function  $f(t)$  along either the vertical or the horizontal axis affects its amplitude–phase Fourier series.
- It is apparent that shifting  $f(t)$  up or down the vertical axis changes only the average value  $K_0 (= A_0)$ , leaving the amplitude  $K_n$  and phase angle  $\phi_n$  of each harmonic component unaffected.



Waveform (b) is obtained by shifting the waveform (a)

# ELEC2134: Circuits and Signals

## Appendix B: Useful Integrals and Trigonometric Identities

# Useful Integrals: These are for guidance only. All answers should be verified by you.

## 1. Integrals (for integer $n \neq 0$ )

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax); \quad \int \cos(ax) dx = \frac{1}{a} \sin(ax)$$

$$\int_0^T \sin(n\omega_0 t) dt = 0; \quad \int_0^T \cos(n\omega_0 t) dt = 0$$

## 2. Integration by Parts (Key Formula) $\boxed{\int u dv = uv - \int v du}$

$$\int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1); \quad \int x^2 e^{ax} dx = \frac{e^{ax}}{a^3} (a^2 x^2 - 2ax + 2)$$

## 3. Integration by Parts: (a) $\int t \cos(n\omega_0 t) dt$

Let  $u = t \Rightarrow du = dt; dv = \cos(n\omega_0 t) dt \Rightarrow v = \frac{1}{n\omega_0} \sin(n\omega_0 t)$

$$\boxed{\int t \cos(n\omega_0 t) dt = \frac{t}{n\omega_0} \sin(n\omega_0 t) - \frac{1}{(n\omega_0)^2} \cos(n\omega_0 t) + G}$$

Constant

## 4. Integration by Parts: (b) $\int t \sin(n\omega_0 t) dt$

Let  $u = t \Rightarrow du = dt; dv = \sin(n\omega_0 t) dt \Rightarrow v = \frac{-1}{n\omega_0} \cos(n\omega_0 t)$

$$\boxed{\int t \sin(n\omega_0 t) dt = -\frac{t}{n\omega_0} \cos(n\omega_0 t) + \frac{1}{(n\omega_0)^2} \sin(n\omega_0 t) + G}$$

## 5. Definite Integration by Parts: For $0 \leq t \leq T$ and integer $n$ :

$$\int_0^T t \sin(n\omega_0 t) dt = \boxed{\frac{T}{n\omega_0}} \quad \text{and} \quad \int_0^T t \cos(n\omega_0 t) dt = \boxed{0}$$

## 6. Definite Integrals

$$\int_0^T x^2 \cos(nx) dx = \left[ \frac{x^2}{n} \sin(nx) + \frac{2x}{n^2} \cos(nx) - \frac{2}{n^3} \sin(nx) \right]_0^T$$

$$\int_0^T x^2 \sin(nx) dx = \left[ \frac{-x^2}{n} \cos(nx) + \frac{2x}{n^2} \sin(nx) + \frac{2}{n^3} \cos(nx) \right]_0^T$$

## 7. Definite Integrals

$$\int_0^T x \cos(nx) dx = \left[ \frac{x}{n} \sin(nx) + \frac{1}{n^2} \cos(nx) \right]_0^T$$

$$\int_0^T x \sin(nx) dx = \left[ \frac{-x}{n} \cos(nx) + \frac{1}{n^2} \sin(nx) \right]_0^T$$

## 8. Definite Integrals

$$\int_0^T \cos(m\omega_0 t) \sin(n\omega_0 t) dt = 0 \quad \text{for all } m \text{ and } n$$

$$\int_0^T \sin(m\omega_0 t) \sin(n\omega_0 t) dt = \begin{cases} 0 & \text{for all } m \neq n \\ \frac{T}{2} & \text{for } m = n \end{cases}$$

$$\int_0^T \cos(m\omega_0 t) \cos(n\omega_0 t) dt = \begin{cases} 0 & \text{for all } m \neq n \\ \frac{T}{2} & \text{for } m = n \end{cases}$$

# Trigonometric Identities: All formulae should be verified by you.

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\cos(\alpha \pm 90^\circ) = \mp \sin \alpha$$

$$\sin(\alpha \pm 90^\circ) = \pm \cos \alpha$$

$$\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta)$$

$$\sin \alpha \sin \beta = \frac{1}{2} \cos(\alpha - \beta) - \frac{1}{2} \cos(\alpha + \beta)$$

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$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

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$$\sin \alpha = \frac{e^{j\alpha} - e^{-j\alpha}}{2j}$$

$$\cos \alpha = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$$

$$e^{\pm j\alpha} = \cos \alpha \pm j \sin \alpha$$

$$A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \cos \left( \alpha + \tan^{-1} \left( \frac{B}{A} \right) \right)$$

## Useful formulae

### 1. Integrals (for integer $n \neq 0$ )

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax); \quad \int \cos(ax) dx = \frac{1}{a} \sin(ax)$$

$$\int_0^T \sin(n\omega_0 t) dt = 0; \quad \int_0^T \cos(n\omega_0 t) dt = 0$$

### 2. Integration by Parts (Key Formula) $\int u dv = uv - \int v du$

$$\int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1); \quad \int x^2 e^{ax} dx = \frac{e^{ax}}{a^3} (a^2 x^2 - 2ax + 2)$$

### 3. Integration by Parts: (a) $\int t \cos(n\omega_0 t) dt$

Let  $u = t \Rightarrow du = dt; dv = \cos(n\omega_0 t) dt \Rightarrow v = \frac{1}{n\omega_0} \sin(n\omega_0 t)$

$$\int t \cos(n\omega_0 t) dt = \frac{t}{n\omega_0} \sin(n\omega_0 t) - \frac{1}{(n\omega_0)^2} \cos(n\omega_0 t) + G$$

Constant

### 4. Integration by Parts: (b) $\int t \sin(n\omega_0 t) dt$

Let  $u = t \Rightarrow du = dt; dv = \sin(n\omega_0 t) dt \Rightarrow v = \frac{-1}{n\omega_0} \cos(n\omega_0 t)$

$$\int t \sin(n\omega_0 t) dt = -\frac{t}{n\omega_0} \cos(n\omega_0 t) + \frac{1}{(n\omega_0)^2} \sin(n\omega_0 t) + G$$

### 5. Definite Integration by Parts: For $0 \leq t \leq T$ and integer $n$ :

$$\int_0^T t \sin(n\omega_0 t) dt = \left[ \frac{T}{n\omega_0} \right] \quad \text{and} \quad \int_0^T t \cos(n\omega_0 t) dt = [0]$$

### 6. Definite Integrals

$$\int_0^T x^2 \cos(nx) dx = \left[ \frac{x^2}{n} \sin(nx) + \frac{2x}{n^2} \cos(nx) - \frac{2}{n^3} \sin(nx) \right]_0^T$$

$$\int_0^T x^2 \sin(nx) dx = \left[ -\frac{x^2}{n} \cos(nx) + \frac{2x}{n^2} \sin(nx) + \frac{2}{n^3} \cos(nx) \right]_0^T$$

### 7. Definite Integrals

$$\int_0^T x \cos(nx) dx = \left[ \frac{x}{n} \sin(nx) + \frac{1}{n^2} \cos(nx) \right]_0^T$$

$$\int_0^T x \sin(nx) dx = \left[ -\frac{x}{n} \cos(nx) + \frac{1}{n^2} \sin(nx) \right]_0^T$$

### 8. Definite Integrals

$$\int_0^T \cos(m\omega_0 t) \sin(n\omega_0 t) dt = 0 \quad \text{for all } m \text{ and } n$$

$$\int_0^T \sin(m\omega_0 t) \sin(n\omega_0 t) dt = \begin{cases} 0 & \text{for all } m \neq n \\ \frac{T}{2} & \text{for } m = n \end{cases}$$

$$\int_0^T \cos(m\omega_0 t) \cos(n\omega_0 t) dt = \begin{cases} 0 & \text{for all } m \neq n \\ \frac{T}{2} & \text{for } m = n \end{cases}$$

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\cos(\alpha \pm 90^\circ) = \mp \sin \alpha$$

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$$A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \cos \left( \alpha + \tan^{-1} \left( \frac{B}{A} \right) \right)$$

## Useful formulae for Fourier series

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t);$$

$$A_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \quad A_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt$$

$$B_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt$$

**Odd function:**  $x(t) = -x(-t)$

$$A_n = 0 \text{ for all } n$$

$$B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$

**Half-wave symmetry:**  $x(t) = -x(t - \frac{T}{2})$

$$A_n = 0 \text{ for even } n; \quad B_n = 0 \text{ for even } n$$

$$A_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \text{ for odd } n$$

$$B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \text{ for odd } n$$

**Even function:**  $x(t) = x(-t)$

$$B_n = 0 \text{ for all } n$$

$$A_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$$

$$x(t) = K_0 + \sum_{n=1}^{\infty} K_n \cos(n\omega_0 t + \phi_n)$$

$$K_n \angle \phi_n = A_n - jB_n;$$

$$K_n = \sqrt{A_n^2 + B_n^2}; \quad \phi_n = -\tan^{-1} \left( \frac{B_n}{A_n} \right)$$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

$$C_n = \frac{1}{2} (A_n - jB_n)$$

For Even Symmetry:

$$C_n = \frac{2}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$$

For Odd Symmetry:

$$C_n = \frac{-2j}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$